

# Advanced Social Text and Media Analytics

## a) Difference between Data Mining and Text Mining.

### Data Mining:

- Data mining is the process of finding patterns and extracting useful data from large data sets.
- Data mining is the process of discovering patterns and knowledge from large amounts of data. It involves the use of various techniques such as machine learning, statistical analysis, and database management to extract insights and information from data.
- Text mining is just a part of data mining.
- Data mining can be applied to a wide range of data types such as *numerical, categorical, and image data*. It can be used for tasks such as prediction, classification, clustering, and association rule mining. Data mining can be applied in various industries such as finance, healthcare, retail, and manufacturing.

### Text Mining:

- Text mining is basically an artificial intelligence technology that involves processing the data from various text documents.
- Text mining, also known as text data mining, is a specific application of data mining that deals with unstructured text data.
- It involves the use of natural language processing (NLP) techniques to extract useful information and insights from large amounts of unstructured text data, such as documents, emails, and social media posts.
- Text mining can be used for tasks such as sentiment analysis, named entity recognition, and topic modeling.
- Text mining can be used in various fields such as natural language processing, information retrieval, and social media analysis.

What exactly is text mining?

The most basic description would be that text mining is *Data mining* over *Text data*. Well ... but what's data mining? Let's put it like this ... Let us imagine you have some raw data which does not give much valuable information

```
{  
  "created_at": "Fri July 23 19:49:24",  
  "id": 102236738912732,  
  "text": "Woah....I never expected this from the Barbie Movie! ....."  
}
```

Here you have the text but it by itself does not give much information to anyone. So basically extracting logical, previously unknown and potentially useful information from data like this can be defined as data mining. To get a better understanding of this process, let us look at a Data, Information, Knowledge, Wisdom (DIKW) Model.

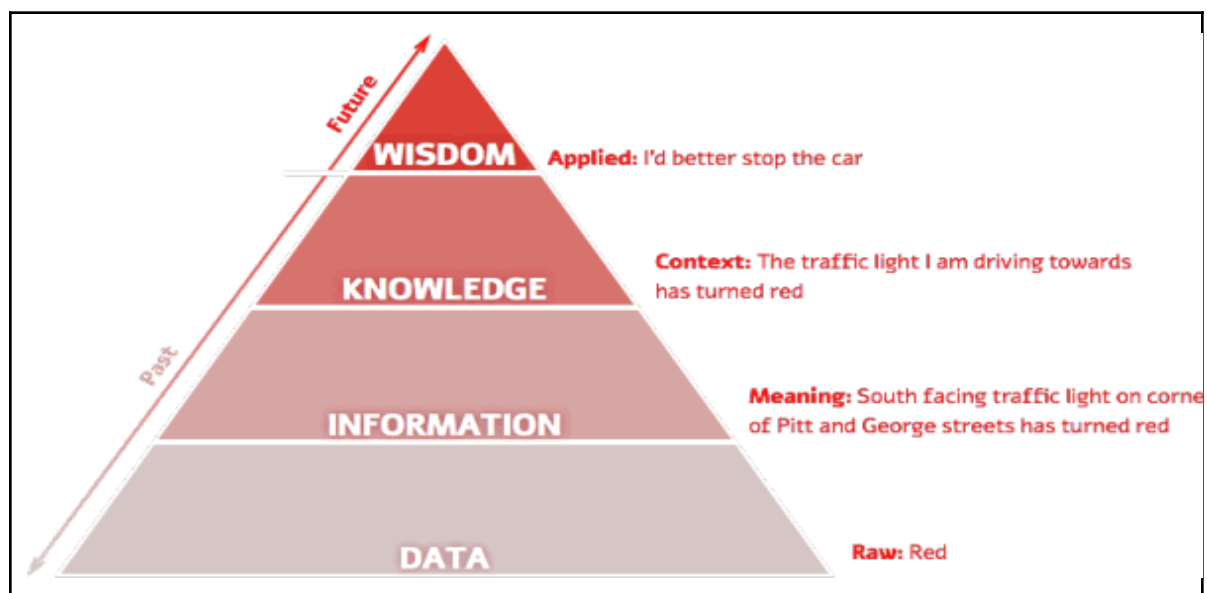
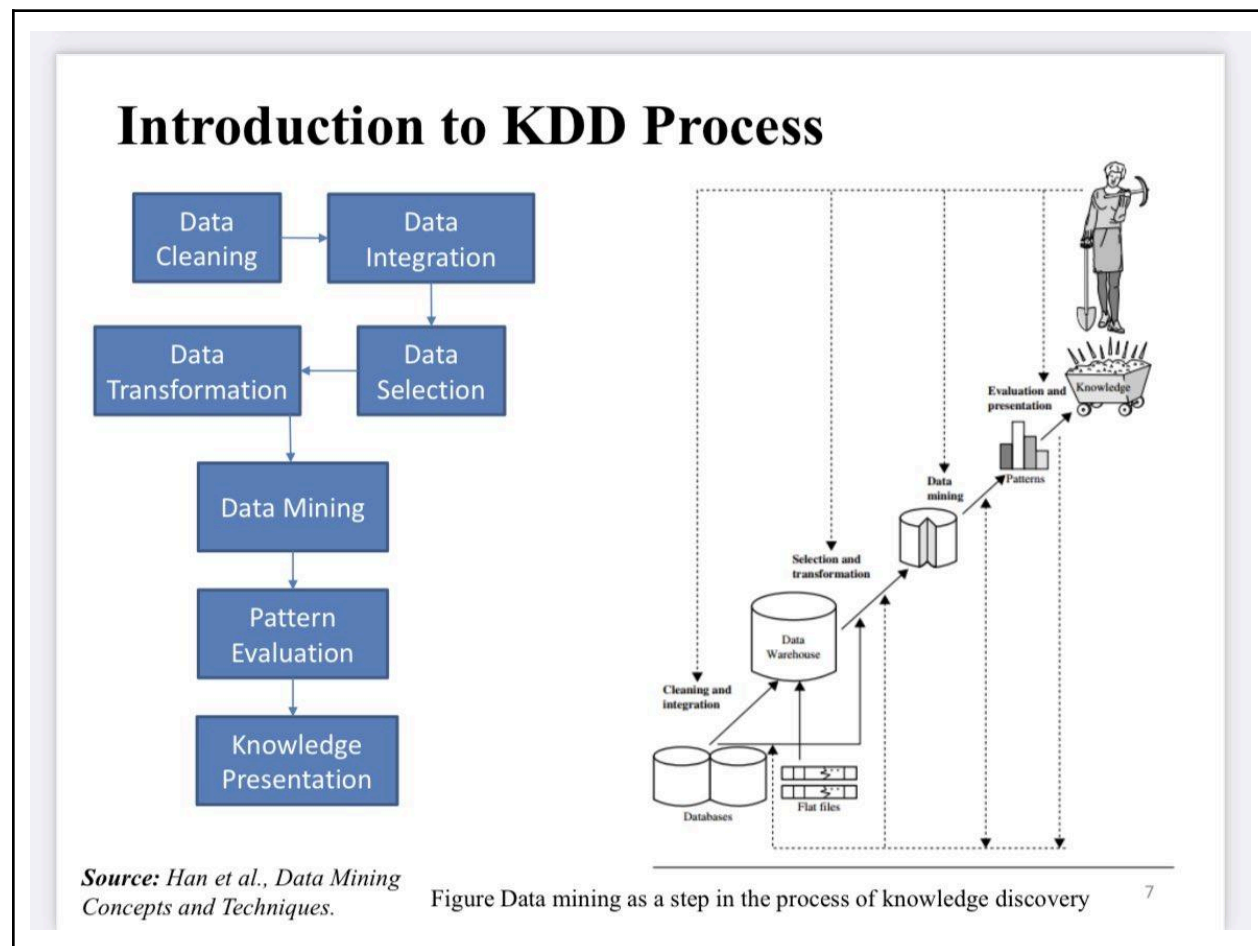


FIG: DIKW model

However, the process of getting from data (text) to knowledge is quite a big process. Data mining or text mining is one of the most important phases of this process.



## What is Text?

Let us first get a basic idea on what text is. When it comes to data itself, there are different types of data as below,

Data, data, data! Today's world is revolving around data. Data backs every decision we make. Therefore, understanding the different types of data is very crucial.

Based on Measurement:

Quantitative Data:

Quantitative data is specific and objective. It can be measured or counted using numbers. For example, you bought a smartphone. The price of the smartphone is an example of quantitative data.

Quantitative data is of two types:

**Continuous Data:**

Continuous data can take any real numeric value. It can be measured, but it cannot be counted. For example, the screen size of your smartphone. It can be anything — from 5 inches to 6.3 inches. One can never say that a particular value within the range is not possible.

***Interval Data:***

Interval Data is ordered data that has equal difference between consecutive observations. It does not have an absolute zero, which means it can dip below zero. For example, temperature (in C or F).

|                                           |
|-------------------------------------------|
| <b>Remember, Quantitative = Quantity.</b> |
|-------------------------------------------|

## **Qualitative Data:**

Qualitative data is subjective and explanatory. It cannot be counted or measured using numbers. For example, the best feature in your smartphone: it can be the battery life or the fingerprint sensor, but it cannot be a number.

Qualitative data is of three types:

### **Nominal Data:**

Nominal data is not categorized in a specific order. For example, the manufacturing company of your smartphone: the name of the company does not specify any order.

### **Ordinal Data:**

Ordinal data has a set order. For example, your smartphone experience: it can be very good, good, bad, or very bad. The use of these terms have a specific order: very good > good > bad > very bad.

### **Binomial Data:**

Binomial data can have only two values. For example, if someone questions whether you like a particular application, you reply in yes or no.

Remember, Qualitative = Quality

## **BASED ON FORMAT:**

### **Types of Data**

#### **1. Well Structured Data :**

- Highly organised and adheres to a predefined data model
- Easy to search, manipulate & analyze data.
- Example — Relational databases, spreadsheets

#### **2. Semi-Structured Data :**

- Combination of structured & unstructured data — No rigid data model
- Example Data formats: XML, JSON, CSV

#### **3. Unstructured Data:**

- No fixed data model
- Examples: Images, Audio, **Text**, Videos

*Text* usually falls under the type of *unstructured* data.

## **Based on Volume:**

### **Big Data:**

- "Big Data" refers to large chunks of structured and unstructured data that has to be divided into parts for processing.
- Since the volume of the data is huge, its quality is not guaranteed. For example, the image or video posts on social media are uploaded on a daily basis.

### **Small Data:**

- Small data refers to data that is small enough for humans to comprehend in terms of size and format.
- It contains much less noise and is collected in a controlled manner. For example, every time a person logs into their social media account, a time stamp is generated.

## **Based on Collection:**

### **Primary Data:**

Primary data is collected from first-hand sources. For example, a survey you conducted. This data is 100% reliable and can be used immediately for analysis purposes.

### **Secondary Data:**

Secondary data is collected by other people and used by you. For example, data obtained from an analytics company. This data is not 100% reliable and has to be verified from different sources before usage.

## Based on Source:

### **Internal Data:**

Internal data originates from and is owned by an organization and lives within its own servers. For example, data on employees from the organization's human resources department.

### **External Data:**

External Data is not owned by the company and is available outside the company's servers. For example, customer feedback.

## Based on Time:

### **Historical Data:**

Historical data is collected and stored over a long period of time. For example, log files and documentation.

### **Real-Time Data:**

Real-Time Data is generated every second and delivered immediately after collection. For example, smart health watches that monitor heart rate.

## Common Sources of Text

Text need for text mining can be obtained through different sources such as,

- Raw Text Documents
- Non-Raw Text documents (DOCX, PDFs)
- Public APIs
- HTML Web pages
- Databases and etc.

## Most Common Text Mining Tasks

There are several common text mining tasks such as,

- Basic pattern/ Sub string matching
- Text Clustering
- Text Classification
- Keyword Extraction
- Text Summarisation
- Named Entity Relationship
- Relationship extraction, etc.

## Text Mining:

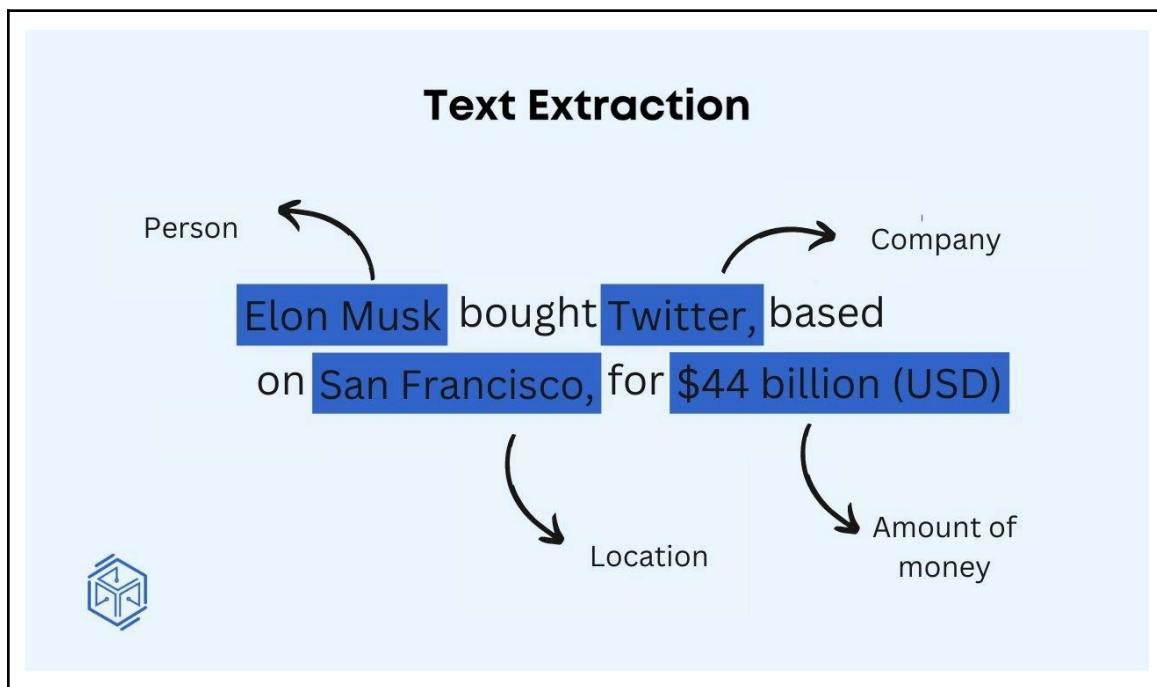
- The extraction of valuable insights from text is known as “text mining,” “text data mining,” or “text analytics,” as defined by Wikipedia.
- This definition nails the heart of text mining, which is to sift through **unstructured information to find the usage patterns and insights** that are necessary for investigating such information.
- Text mining can be accurately described as “multidisciplinary.” These disciplines include information retrieval, data mining, machine learning, statistics, and computational linguistics.
- Text mining is the process of extracting useful information from texts written in natural languages and stored in either semi-structured or unstructured formats.

## Text Mining Techniques

To comprehend text mining methods, one must look into the steps taken to mine text and deduce meaning. The majority of these text-mining strategies require the use of various text-mining software programs.

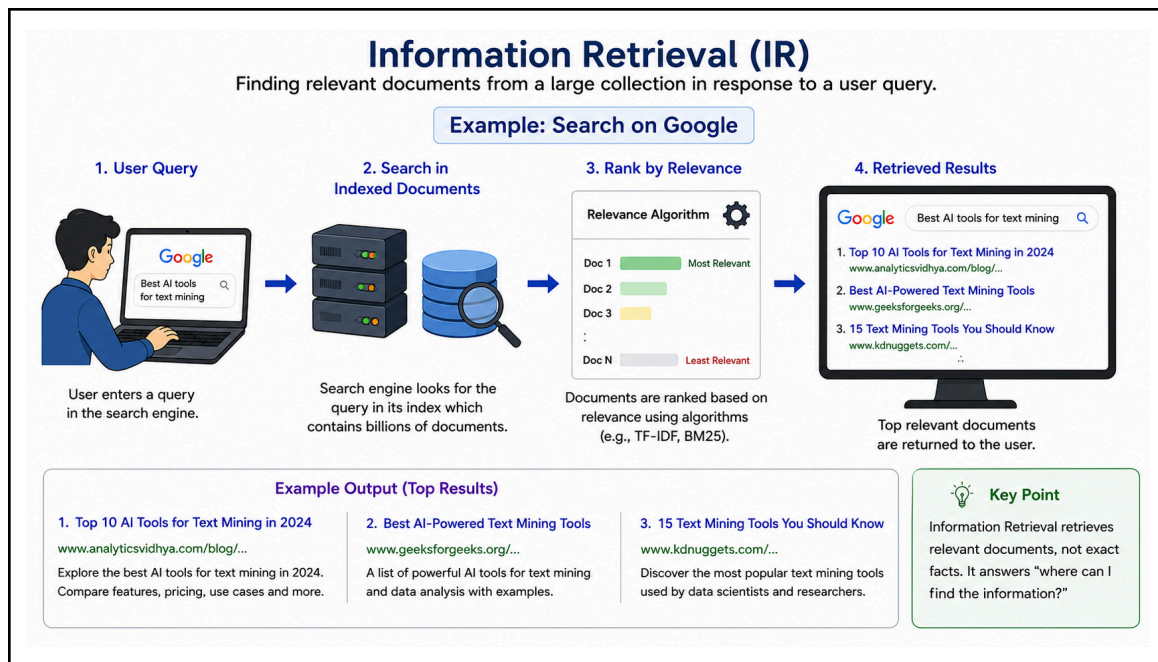
## Information Extraction

- In the world of text mining, this is the gold standard. What we mean when we talk about “information exchange” is the procedure of gleaning relevant details from massive swaths of textual data.
- This text mining method seeks to extract entities, properties, and relationships from unstructured or semi-structured texts by analyzing their content.
- A database is used to permanently store the extracted data for later use.
- Precision and recall techniques are used to assess and evaluate the efficacy and relevance of the results. Information Extraction is a valuable method for studying textual material.



# Information Retrieval

- The term “information retrieval” (or “IR”) is used to describe the method used to find data that fits a certain description or criteria based on a keyword or combination of keywords.
- Using a variety of techniques, IR systems may now mine texts for useful information by keeping tabs on user actions.
- Both Google and Yahoo are among the most well-known search engines, making them examples of IR systems.
- Information Retrieval (IR) is the most well-known method employed in text mining



Suppose a student searches:

"Machine Learning notes PDF" on search engine.

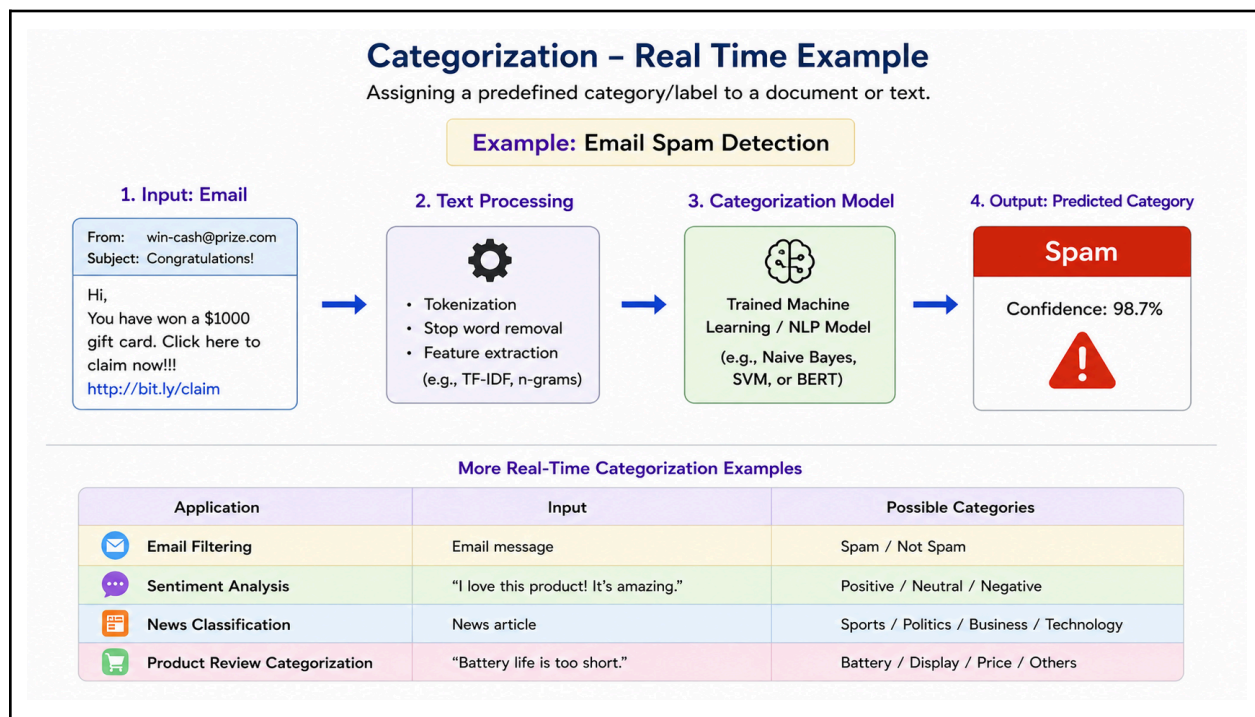
- Information retrieval returns:
  - PDF lecture notes
  - Research papers
  - Tutorial websites
  - YouTube videos
- Information Extraction would then read one of those PDFs and extract:
  - Topic: Machine Learning
  - Algorithms: Decision Tree, SVM, Neural Networks
  - Author: John Smith
  - Publication Year: 2025

In one sentence:

Information Retrieval answers "**Which documents are relevant?**" while Information Extraction answers "**What important facts are inside those documents?**"

## Categorization:

- This is an example of a “supervised” learning method used in text mining, in which texts written in a natural language are classified as determined by the themes they cover.
- Therefore, classification, or more precisely, Natural Language Processing (NLP), is the act of collecting text documents and analyzing them to unearth the appropriate subjects or indexes for each document.
- Categorization is an effective method for studying written records.



## Clustering

In text mining, clustering is an unsupervised learning technique where documents are grouped together based on similarity of content, without predefined labels.

- No prior categories are given
- The system automatically finds patterns
- Similar documents are placed in the same group (cluster)
- When it comes to text mining, clustering is a must-have strategy.

Specifically, it aims to find underlying patterns in textual data and classify those patterns into meaningful clusters.

- Unlabeled textual data presents a big problem in the clustering process since it is difficult to build meaningful clusters without any prior information on them.

## **Scenario: E-commerce customer reviews (Amazon/Flipkart)**

### **A) Clustering (No labels given)**

A company has 1 lakh customer reviews:

“Battery drains fast”

“Camera quality is poor”

“Phone heats up quickly”

“Excellent display colors”

The system automatically groups them:

Cluster 1 → Battery issues

Cluster 2 → Camera issues

Cluster 3 → Performance issues

Cluster 4 → Positive feedback

### **B) Text Categorization (Labels given)**

Now assume the company already defined categories:

- Battery Issue
- Camera Issue
- Performance Issue
- Positive Review

Model is trained and then classifies: “Phone gets heated while gaming”

Performance Issue (predicted directly)

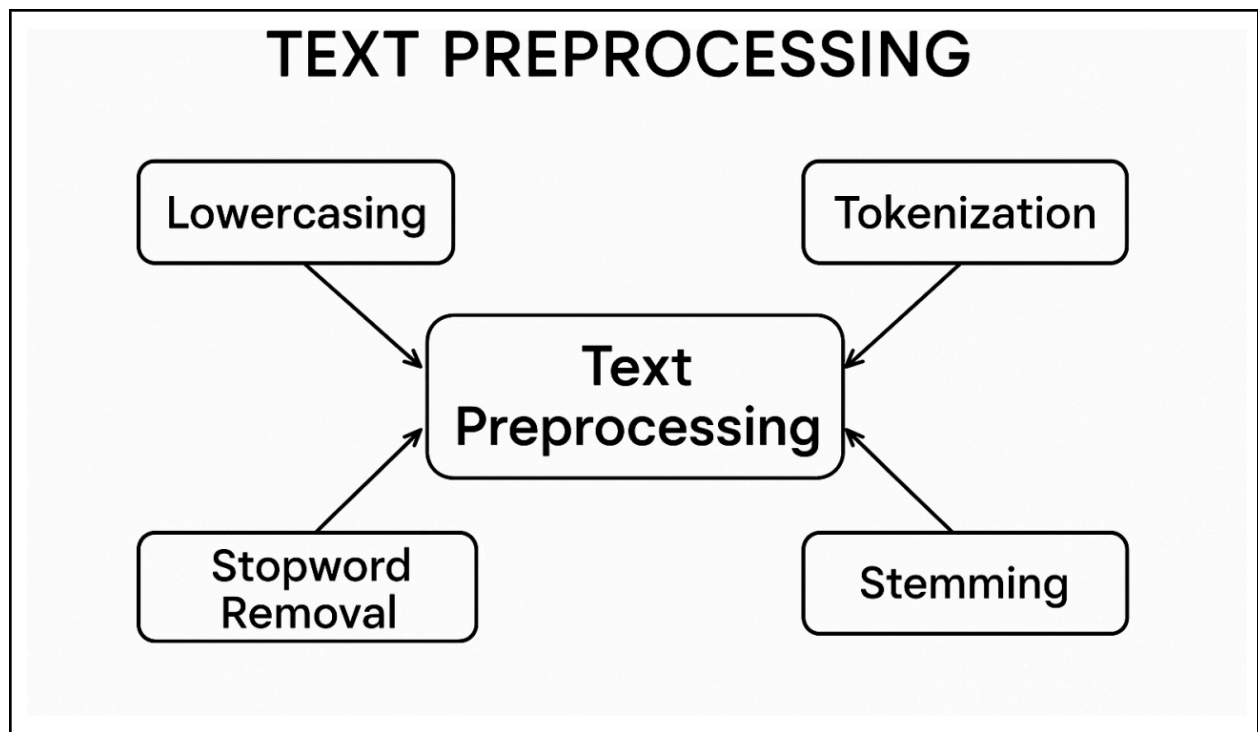
## **Summarisation**

- The term “text summarisation” describes the method of automatically producing a condensed version of a certain text that contains useful information for the end user.
- The goal of this text-mining technique is to examine a large number of textual resources and produce condensed versions of longer texts that retain the core content and purpose of the originals while presenting the data in a more manageable fashion.
- The process of text summarization combines and integrates the different techniques that use text categorization, such as decision trees, neural networks, regression models, and swarm intelligence.

## Why is text preprocessing important?

Data quality significantly influences the performance of a machine-learning model. Inadequate or low-quality data can lead to lower accuracy and effectiveness of the model.

In general, text data derived from natural language is unstructured and noisy. So text preprocessing is a critical step to transform messy, unstructured text data into a form that can be effectively used to train machine learning models, leading to better results and insights.



Text preprocessing refers to a series of techniques used to clean, transform and prepare raw textual data into a format that is suitable for NLP or ML tasks. The goal of text preprocessing is to enhance the quality and usability of the text data for subsequent analysis or modeling.

- Lowercasing
- Removing Punctuation & Special Characters
- Stopword Removal
- Removal of URLs
- Removal of HTML Tags
- Stemming & Lemmatization
- Tokenization

Some or all of these text preprocessing techniques are commonly used by NLP systems. The order in which these techniques are applied may vary depending on the needs of the project

**Lowercasing:** Lowercasing is a text preprocessing step where all letters in the text are converted to lowercase. This step is implemented so that the algorithm does not treat the same words differently in different situations.

```
text = "Hello World!"
lowercased_text = text.lower()

print(lowercased_text)
```

Output:  
hello world!

**Removing Punctuation & Special Characters:** Punctuation removal is a text preprocessing step where you remove all punctuation marks (such as periods, commas, exclamation marks, emojis etc.) from the text to simplify it and focus on the words themselves.

```
import re

text = "Hello, world! This is? * 🍷 an&/|~^+%'\" example- of text preprocessing."

punctuation_pattern = r'[^\\w\\s] '

text_cleaned = re.sub(punctuation_pattern, '', text)

print(text_cleaned)
```

Output:  
Hello world This is an example of text preprocessing

## Stop-Words Removal

Stopwords are words that don't contribute to the meaning of a sentence. So they can be removed without causing any change in the meaning of the sentence. The NLTK library has a set of stopwords and we can use these to remove stopwords from our text and return a list of word tokens. Removing these can help focus on the important words

**Reduces Data Dimensionality:** By dropping these words, you significantly decrease the size of your text data and vocabulary matrix, saving computational storage and memory.

**Improves Signal-to-Noise Ratio:** Machine learning models (like text classifiers) can better identify the core topics or intent of a document when uninformative filler words are removed

## Removal of URLs

This preprocessing step is to remove any URLs present in the data

```
def remove_urls(text):  
    url_pattern = re.compile(r'https?://\S+|www\.\S+')  
    return url_pattern.sub(r'', text)  
  
text = "I hope it will be a useful article for you. Follow me:  
https://medium.com/@ayselaydin"  
remove_urls(text)  
  
Output:  
I hope it will be a useful article for you. Follow me:
```

## Why Remove URLs in NLP?

- **Noise Reduction:** Hyperlinks do not carry semantic linguistic meaning.
- **Model Accuracy:** Leaving URLs in the text can confuse machine learning models, leading to skewed results in **sentiment analysis** or **text classification**.
- **Dataset Streamlining:** Cleans up social media or web scraping data, allowing the model to focus on meaningful text patterns.

## Removal of HTML Tags

Removal of HTML Tags is a text preprocessing step used to clean text data from HTML documents. When working with text data obtained from web pages or other HTML-formatted sources, the text may contain HTML tags, which are not desirable for text analysis or machine learning models. Therefore, it's important to remove HTML tags from the text data

```
import re

text = """<html><div>
<h1>Aysel Aydin</h1>
<p>Text Preprocessing for NLP</p>
<a href="https://medium.com/@ayselaydin">Medium account</a>
</div></html>"""

html_tags_pattern = r'<.*?>'

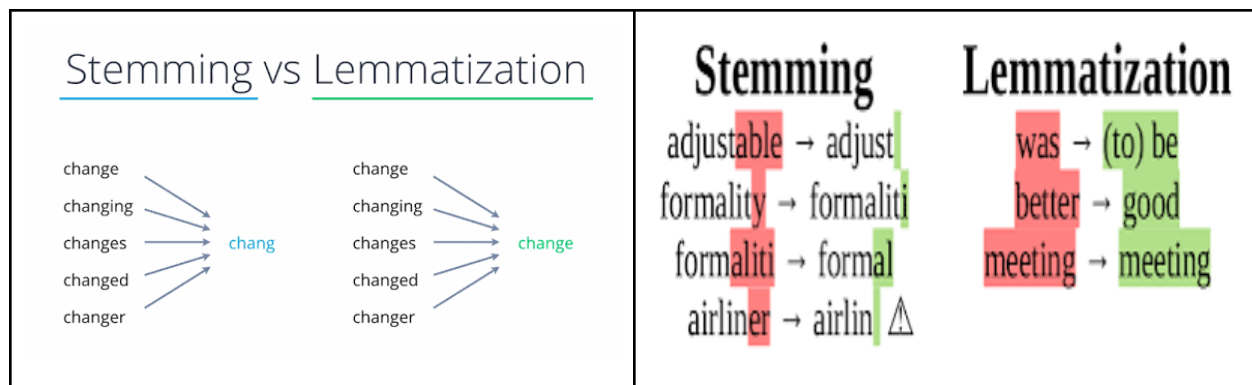
text_without_html_tags = re.sub(html_tags_pattern, '', text)

print(text_without_html_tags)
```

Output:  
Aysel Aydin  
Text Preprocessing for NLP  
Medium account

## Stemming & Lemmatisation:

- Stemming and lemmatisation are two text preprocessing techniques used to reduce words to their base or root form. The primary goal of these techniques is to reduce the number of unique words in a text document, making it easier to analyze and understand.
- They are widely used for Search engines and tagging. Search engines use stemming for indexing the words. Therefore, instead of storing all forms of a word, a search engine may only store its roots. In this way, stemming reduces the size of the index and increases retrieval accuracy.
- Stemming involves removing suffixes from words to obtain their base form while lemmatization involves converting words to their morphological base form.
- Stemming is a **simpler** and **faster** technique compared to lemmatization. It uses a set of rules or algorithms to remove suffixes and obtain the base form of a word. However, stemming can sometimes produce a base form that is not valid, in which case it can also lead to ambiguity.
- On the other hand, lemmatization is a more sophisticated technique that uses vocabulary and morphological analysis to determine the base form of a word. Lemmatization is **slower** and **more complex** than stemming. It produces a valid base form that can be found in a dictionary, making it more accurate than stemming.



Tokenization: Natural Language Processing (NLP) has rapidly evolved in recent years, enabling machines to understand and process human language. At the core of any NLP pipeline lies tokenization, a fundamental step that breaks down unstructured text into discrete elements

## Why do we need Tokenization?

Unstructured text data, such as articles, social media posts, or emails, lacks a predefined structure that machines can readily interpret. Tokenization bridges this gap by breaking down the text into smaller units called tokens. These tokens can be words, characters, or even subwords, depending on the chosen tokenization strategy. By transforming unstructured text into a structured format, tokenization lays the foundation for further analysis and processing.

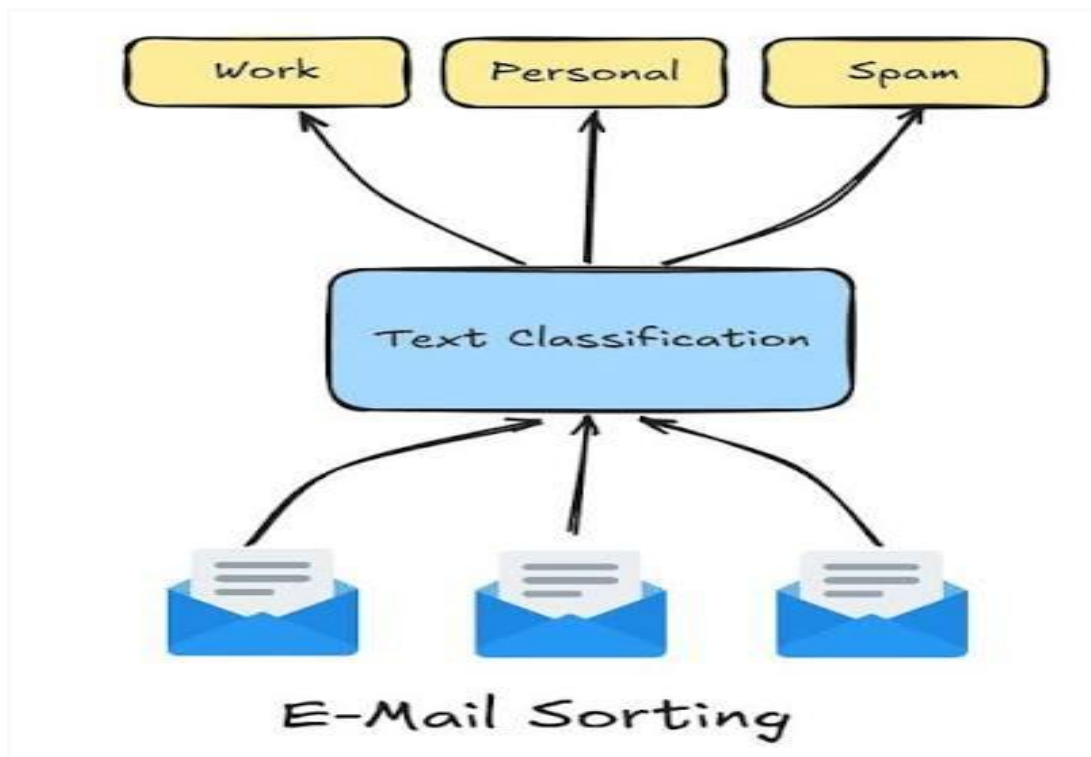
One of the primary reasons for tokenization is to convert textual data into a numerical representation that can be processed by machine learning algorithms. With this numeric representation we can train the model to

perform various tasks, such as classification, sentiment analysis, or language generation.

Various kinds of tokenisations, such as character tokenisation, word tokenisation, sub-word tokenisation also exist.

## Text Categorization & Text Clustering

- Text classification is a process to teach machines to automatically categorize pieces of text into predefined categories or classes.
- Think of it like having a smart assistant that can sort your emails into "work," "personal," and "spam" folders, or a system that can determine whether a movie review is positive or negative.



- Machines actually "read" and understand text, which is quite different from how humans do it. Unlike humans, machines can not naturally understand words and their meanings.
- Machines work with numbers, not text. Therefore, human language is transformed into a format that machines can process mathematically. This is done by converting words into numbers.
- Imagine you're teaching a computer to understand text the way you might teach a child to understand a new language using building blocks. The first step is to break down text into smaller pieces.
- Words are converted to numbers through various methods. One simple approach is "**one-hot encoding**," where each word gets its own unique number or vector.
- More advanced methods like "**word embeddings**" represent words as points in a multi-dimensional space, where similar words are closer together. For example, in a basic number system, the sentence "I love pizza" might become something like [4, 12, 8], where each number represents a word.
- Once text is converted to numbers, machines can start recognising patterns. It learns that certain number combinations (representing words) often appear together in specific categories. For example, in restaurant reviews, positive reviews might often contain number patterns representing words like "delicious," "excellent," "amazing" and negative reviews might show patterns representing "disappointing," "cold," "poor".

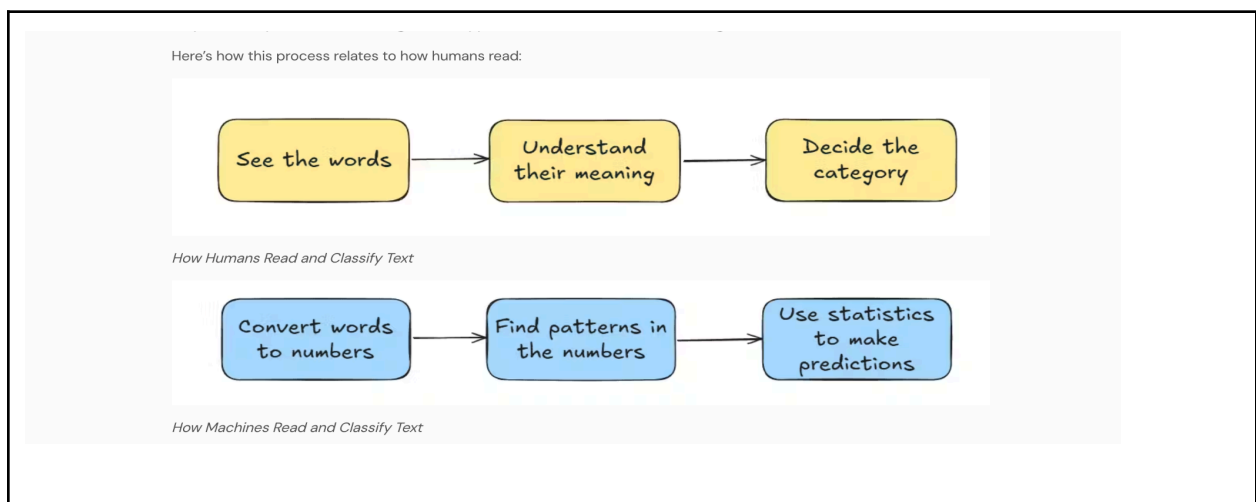
Machines also learn the order of words and the meaning of combinations. For better understanding, they break down the following:

- **Word order:** "The dog chased the cat" is different from "The cat chased the dog"
- **Context:** The word "bank" means something different in "river bank" versus "bank account"
- **Relationships:** Understanding that "excellent" and "outstanding" are similar in meaning

Finally, the machine uses this processed information to make classification decisions. It's similar to how you might recognize a song's genre by picking up on certain patterns of instruments, rhythm, and style. For example, If the a machine sees a new sentence like:

"The weather today is sunny and warm."

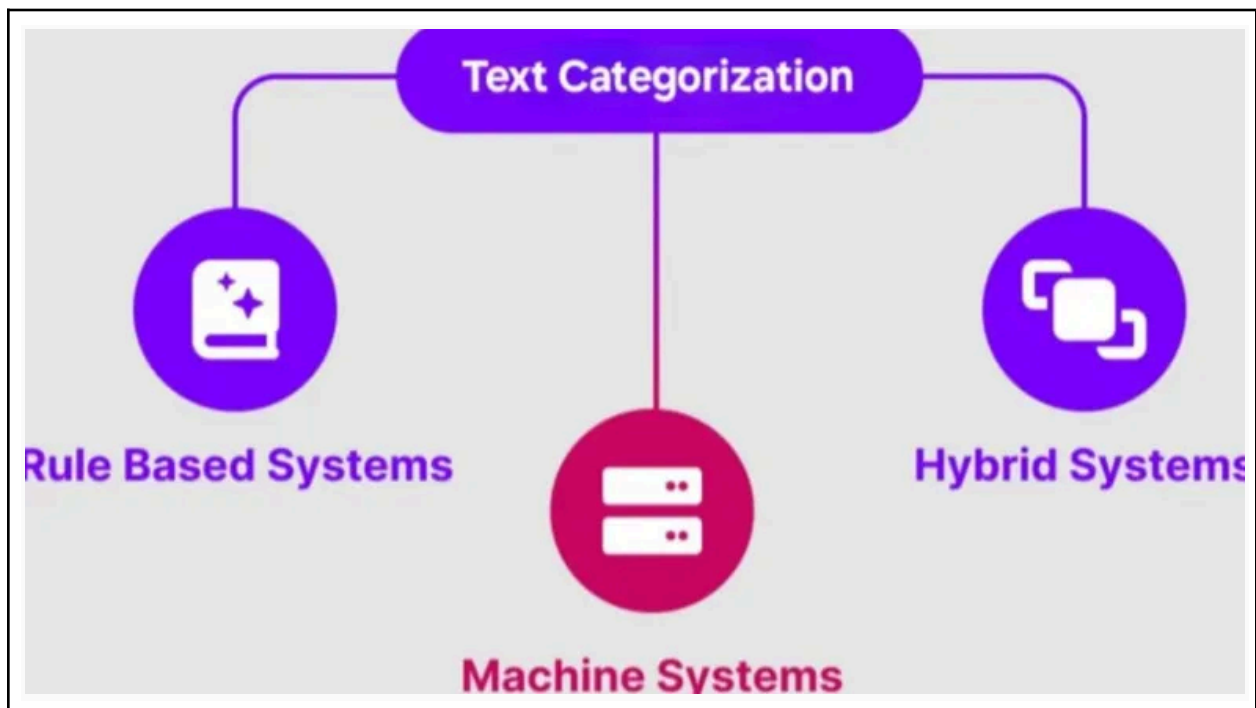
It might classify it as Sunny Weather because it recognizes patterns from previous examples. While machines process text very differently from humans, the goal is to approximate human-like understanding.



The main difference is that humans naturally understand meaning, while machines rely on patterns and probabilities.

## The Evolution of Text Classification Techniques

Over the years, various methods have been developed for text classification. These methods or techniques range from traditional machine learning algorithms to advanced deep learning techniques. Let's look at some of these methods:



### Rule-Based Methods

- Rule-based methods are one of the oldest and most intuitive approaches to text classification.

- These systems rely on manually crafted linguistic rules that are specifically designed to identify patterns or characteristics within the text and assign predefined categories.
- Despite being traditional, they remain relevant in certain contexts where domain-specific knowledge and interpretability are critical.
- Rule-based methods classify text by applying logical conditions, often written as if-then rules. These rules use features such as:

- **Keywords or Phrases:** Specific words or combinations of words that indicate a category. Example: Emails containing words like "win", "lottery", or "prize" might be classified as spam.

- **Regular Expressions:** Patterns to detect variations of text. Example: Identifying email addresses or phone numbers.

- **Linguistic Features:** Syntax, parts of speech, or other linguistic markers. Example: A sentence starting with “Dear” could indicate a formal letter.

## Traditional Machine Learning Algorithms

Traditional machine learning algorithms are a cornerstone of text classification. Unlike rule-based methods, these algorithms learn patterns from labelled data, making them more scalable and adaptable to diverse tasks. Below is an explanation of some of the most widely used traditional algorithms for text classification.

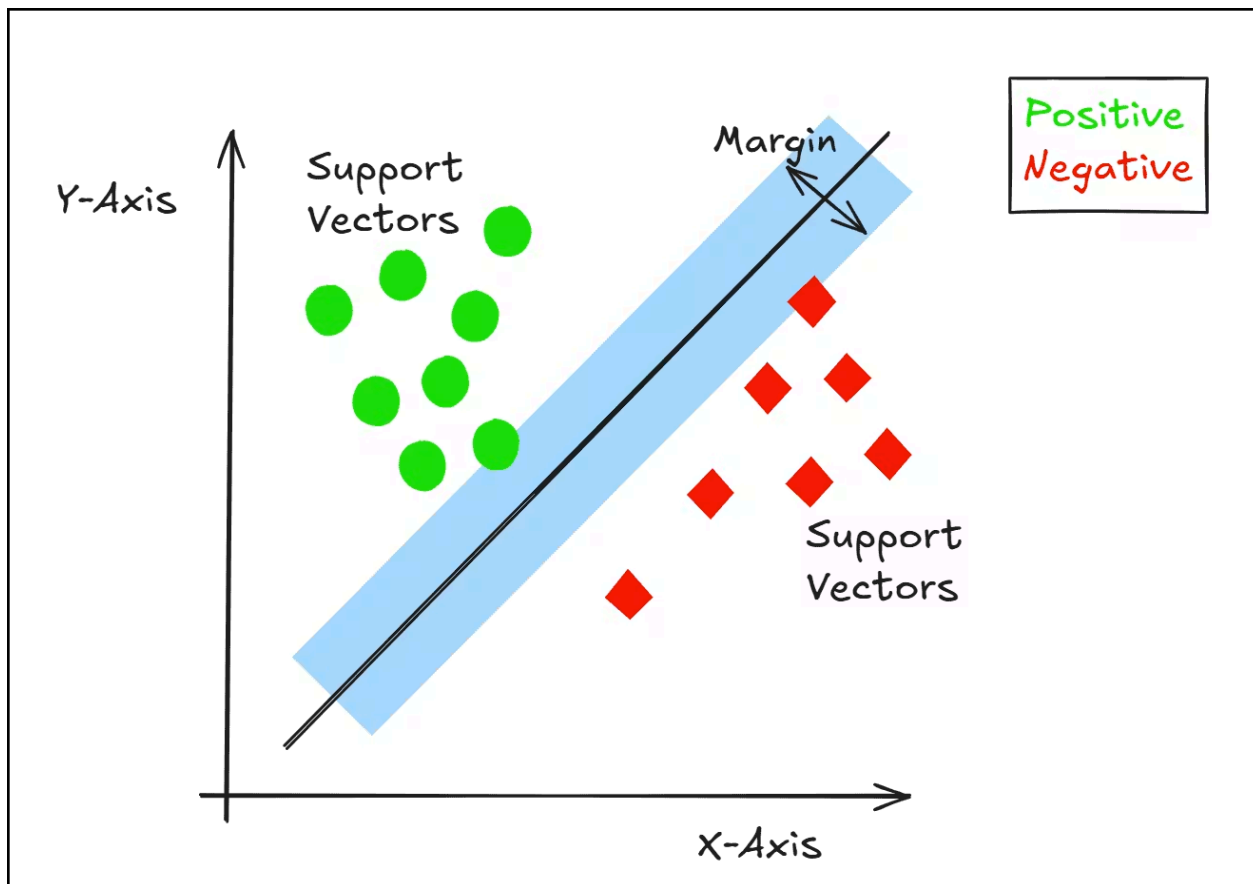
**Naive Bayes Classifier :** is a probabilistic classifier based on Bayes’ Theorem. It assumes that features (words, in text classification) are independent of each other—a "naive"

assumption, hence the name. Despite this assumption, it performs well in many real-world scenarios. Calculates the probability of a text belonging to a class using:

$$P(Class|Text) = \frac{P(Text|Class) \cdot P(Class)}{P(Text)}$$

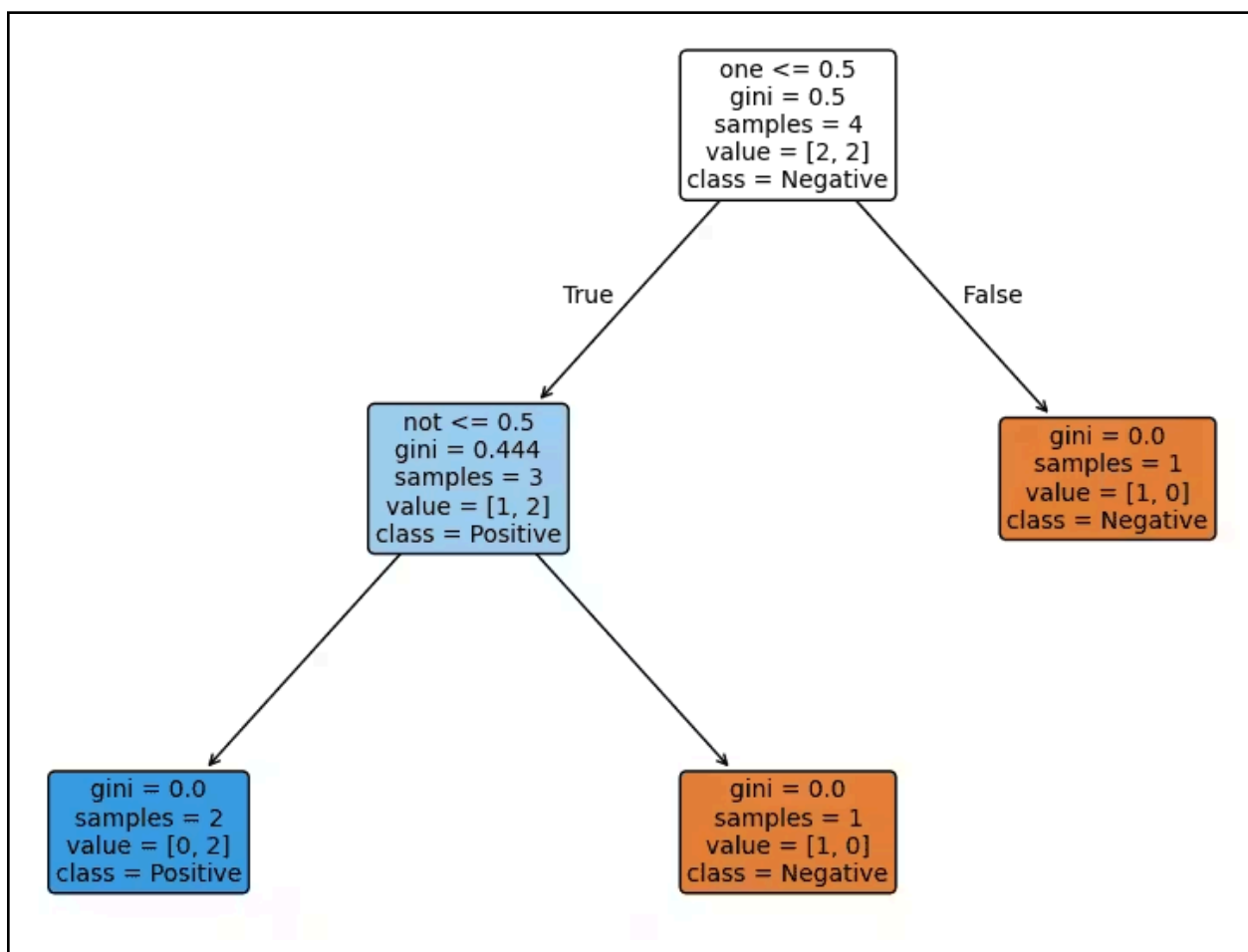
The class with the highest probability is chosen as the predicted category.

**Support Vector Machines (SVM)** is a powerful supervised learning algorithm that finds the best boundary (hyperplane) to separate classes in a high-dimensional space. It works well with sparse datasets like text. SVM maximizes the margin between data points of different classes and the decision boundary. SVM can handle non-linear relationships using kernels .



The above figure shows how SVM separates two classes (Positive and Negative) by finding the optimal hyperplane (black line) that maximizes the margin (blue region) between the closest data points of both classes, called support vectors.

**Decision Trees** :classify data by splitting it based on feature values in a hierarchical manner. The structure resembles a tree where each internal node represents a feature, branches represent decisions, and leaf nodes represent categories. Splits data recursively based on features that maximize information gain or reduce entropy (using criteria like Gini Index or Information Gain). Classification follows the path from the root node to a leaf node.



*Text Representation of the Decision Tree for Positive and Negative classes*

In the above figure, the decision tree predicts sentiment (Positive or Negative) based on the presence of specific words in the text. It evaluates whether words like "one" and "not" appear in the text and uses these conditions to classify the sentiment.

## Teaching Machines to Read and Classify Text

Text classification is a task in NLP where machines are trained to assign predefined categories to pieces of text. It plays a critical role in tasks like sentiment analysis, spam detection, topic categorization, and intent detection in conversational AI.

### Key Components of Text Classification Systems

- **Text Input:** The system processes raw text such as sentences, paragraphs, or entire documents.
- **Preprocessing:** Text is cleaned, tokenised, and converted into numerical representations (embeddings) that models can understand.
- **Modeling:** A machine learning model, often based on transformers like [BERT](#) or [DistilBERT](#), learns patterns and relationships in the text to classify it into one or more categories.
- **Output:** The system outputs a category label or probability distribution over multiple categories.

Here's a simple example of how to train a text classification model using Transformers in a Google Colab notebook. We'll use the Hugging Face transformers library, which provides a user-friendly interface for working with transformer models like BERT. Following are the steps:

- Import the required libraries.
- Load a pre-trained transformer model.
- Use a small dataset (e.g., the IMDB dataset for sentiment analysis).
- Fine-tune the model for text classification.

Questions:

- 1) What is text classification?
- 2) How do machines "understand" text?
- 3) What are the common techniques used in text classification?
- 4) Differentiate between One hot ending and Word embedding in detail.
- 5) What are some practical applications of text classification? Illustrate with the help of a case study.

**Text Clustering:** Text Clustering is an unsupervised machine learning technique that automatically groups similar text documents based on their content without using predefined labels. It helps discover hidden themes and patterns in large collections of text.

Examples

- Customer reviews
- News articles
- Research papers
- Social media posts
- Support tickets

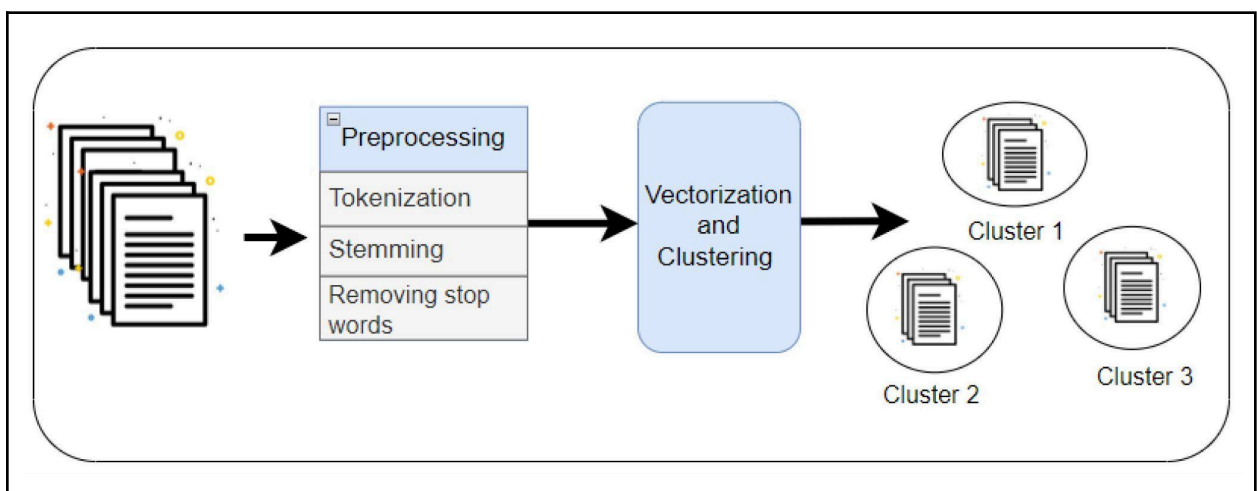
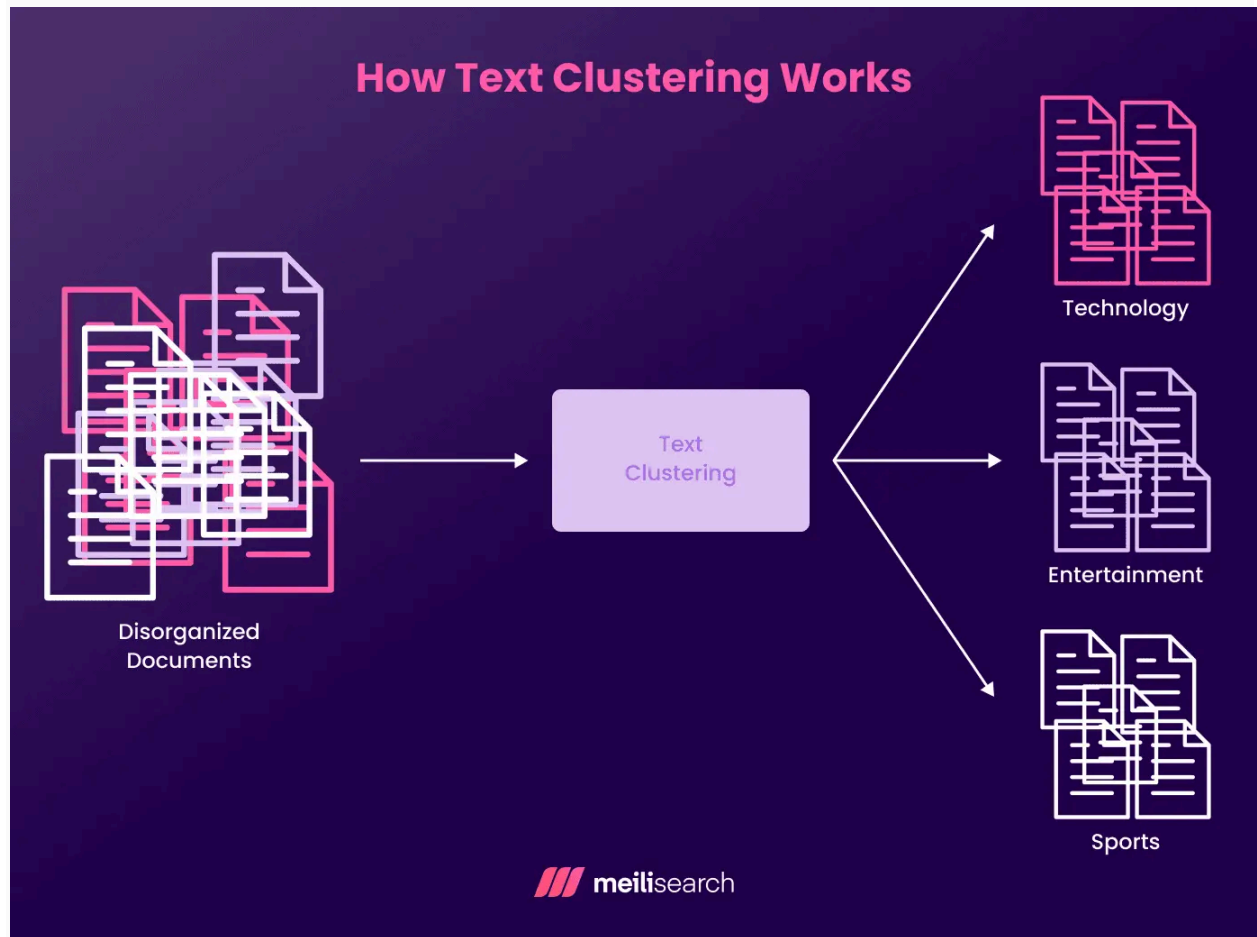
## Real-Time Example

Suppose an e-commerce company receives 10,000 customer reviews.

| Review                        | Cluster             |
|-------------------------------|---------------------|
| "Battery lasts all day"       | Battery Performance |
| "Battery backup is excellent" | Battery Performance |
| "Camera quality is amazing"   | Camera Quality      |
| "Photos are very sharp"       | Camera Quality      |
| "Phone heats while gaming"    | Performance Issues  |

The system automatically groups similar reviews together.

## How Text Clustering Works ?



## Step 1: Text Preprocessing

Raw text is cleaned by:

- Removing punctuation
- Removing URLs
- Removing stop words (the, is, a, an)
- Tokenization
- Stemming/Lemmatization

### Example

Original Text:

"The students are studying machine learning technologies."

After preprocessing:

student study machine learning technology

## Step 2: Text Representation (Embedding)

Text must be converted into numbers.

Common methods:

### 1. TF-IDF

TF-IDF (Term Frequency-Inverse Document Frequency) is a numerical statistic used in NLP and information retrieval to measure a word's relevance to a specific document within a larger collection (corpus). It penalizes common words and rewards rare, meaningful terms.

## How It Works

TF-IDF is the product of two metrics: [\[1\]](#)

$$\text{TF}(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

$$\text{IDF}(t, D) = \log \frac{\text{Total number of documents in corpus } D}{\text{Number of documents containing term } t}$$

This balance allows TF-IDF to highlight terms that are both frequent within a specific document and distinctive across the text document, making it a useful tool for tasks like search ranking, text classification and keyword extraction.

## Converting Text into vectors with TF-IDF

Let's take an example where we have a corpus (a collection of documents) with three documents and our goal is to calculate the TF-IDF score for specific terms in these documents.

1. **Document 1:** "The cat sat on the mat."
2. **Document 2:** "The dog played in the park."
3. **Document 3:** "Cats and dogs are great pets."

Our goal is to calculate the TF-IDF score for specific terms in these documents. Let's focus on the word **"cat"** and see how TF-IDF evaluates its importance.

## Step 1: Calculate Term Frequency (TF)

### For Document 1:

- The word "cat" appears 1 time.
- The total number of terms in Document 1 is 6 ("the", "cat", "sat", "on", "the", "mat").
- So,  $TF(cat, Document\ 1) = 1/6$

### For Document 2:

- The word "cat" does not appear.
- So,  $TF(cat, Document\ 2) = 0$ .

### For Document 3:

- The word "cat" appears 1 time.
- The total number of terms in Document 3 is **6** ("cats", "and", "dogs", "are", "great", "pets").
- So  $TF(cat, Document\ 3) = 1/6$

In Document 1 and Document 3 the word "cat" has the same TF score. This means it appears with the same relative frequency in both documents. In Document 2 the TF score is 0 because the word "cat" does not appear.

## Step 2: Calculate Inverse Document Frequency (IDF)

2. Total number of documents in the corpus (D): 3
3. Number of documents containing the term "cat": 2 (Document 1 and Document 3).

$$IDF(cat, D) = \log \approx 0.176$$

## Step 3: Calculate TF-IDF

The TF-IDF score for "cat" is 0.029 in Document 1 and Document 3 and 0 in Document 2 that reflects both the frequency of the term in the document (TF) and its rarity across the corpus (IDF).

The TF-IDF score is the product of TF and IDF:

$$TF - IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$

- For Document 1:  $TF-IDF(cat, Document\ 1, D) = 0.167 \times 0.176 = 0.029$
- For Document 2:  $TF-IDF(cat, Document\ 2, D) = 0 \times 0.176 = 0$
- For Document 3:  $TF-IDF(cat, Document\ 3, D) = 0.167 \times 0.176 \sim 0.029$

### 3. Word2Vec

**Word2Vec** is a foundational Natural Language Processing (NLP) technique developed by Google. It converts words into numerical vectors (lists of numbers). This allows machines to understand language by placing words with similar meanings close together in a mathematical space

#### How it Works

Instead of random guesses, Word2Vec uses two main shallow neural network architectures to learn word associations from large text datasets:

**CBOW (Continuous Bag of Words):** Predicts a target word based on its surrounding context words.

**Skip-gram:** Takes a target word and predicts the probability of its surrounding context words

#### Common Applications

- **Sentiment Analysis:** Understanding if a review or text is positive or negative based on related word clusters.
- **Search and Recommendation:** Grouping articles, products, or documents with similar context.
- **Translation:** Aligning vector spaces of different languages for initial machine translation matchings

4. BERT Embeddings

5. Transformer Embeddings

| Word             | Vector Representation |
|------------------|-----------------------|
| AI               | [0.2,0.5,0.8]         |
| Machine Learning | [0.3,0.6,0.7]         |

Example: These vectors capture semantic meaning.

### Step 3: Clustering

A clustering algorithm groups similar vectors together.

#### Input

Documents:

- D1: "Artificial Intelligence"
- D2: "Machine Learning"
- D3: "Football Match"
- D4: "Cricket Tournament"

#### Output

Cluster 1:

- D1
- D2

Cluster 2:

- D3

- D4

## Step 4: Evaluation

Measures cluster quality.

Common Metrics:

- Silhouette Score
- Adjusted Rand Index (ARI)
- Davies-Bouldin Index

Higher scores generally indicate better clustering quality

## Popular Text Clustering Algorithms:

- K-Means
- DBSCAN
- Hierarchical Clustering
- Spectral Clustering
- Gaussian Mixture Model (GMM)

# 5. Benefits of Text Clustering

## 1. Automated Organization

No manual labeling required.

## 2. Hidden Pattern Discovery

Finds trends and themes.

### **3. Scalable Analysis**

Handles millions of documents.

### **4. Better Search**

Improves search relevance.

### **5. Data-Driven Insights**

Supports business decisions.

## **6.Applications of Text Clustering**

| <b>Domain</b> | <b>Application</b>          |
|---------------|-----------------------------|
| E-Commerce    | Customer Review Analysis    |
| Education     | Student Feedback Analysis   |
| Healthcare    | Clinical Report Grouping    |
| Research      | Research Paper Organization |

| <b>Domain</b>    | <b>Application</b>  |
|------------------|---------------------|
| News Media       | News Categorization |
| Social Media     | Trend Detection     |
| Customer Support | Ticket Analysis     |

## **7.Challenges in Text Clustering**

### **High Dimensionality**

Thousands of features may exist.

### **Ambiguous Language**

Sarcasm and slang are difficult.

### **Scalability**

Large datasets need significant computation.

## **Choosing Number of Clusters**

Often requires experimentation.

## What is information extraction?

- Information extraction (IE) is the automated process of extracting structured information from semi-structured or unstructured text data, transforming human language text sources such as PDFs into a format that's organized, searchable and machine-readable.
- (NLP) relies on information extraction to identify important data within input text.
- Information extraction algorithms can identify entities, including names, relationships, events, sentiment and more, then classify and store them in a database for further use.
- The resulting structured information has a standardized format and is typically stored in rows and columns that identify its attributes. The standardized storage is the primary differentiator between structured data and unstructured data.
- All the data values within the same database adhere to the same structured format with the same defined attributes.

## Why is information extraction important?

Information extraction allows enterprises to transform documents into actionable datasets and generate valuable insights from them. The

intelligent document processing market—which IE facilitates—is projected to grow at a compound annual growth rate (CAGR) of 33.1% through 2030 from a value of USD 2.3 billion in 2024.

## Types of information extraction

Information extraction tasks are categorized based on the type of information being identified and labeled. IE systems can handle tasks including:

- Named entity recognition (NER)
- Relation extraction
- Event extraction
- Sentiment analysis

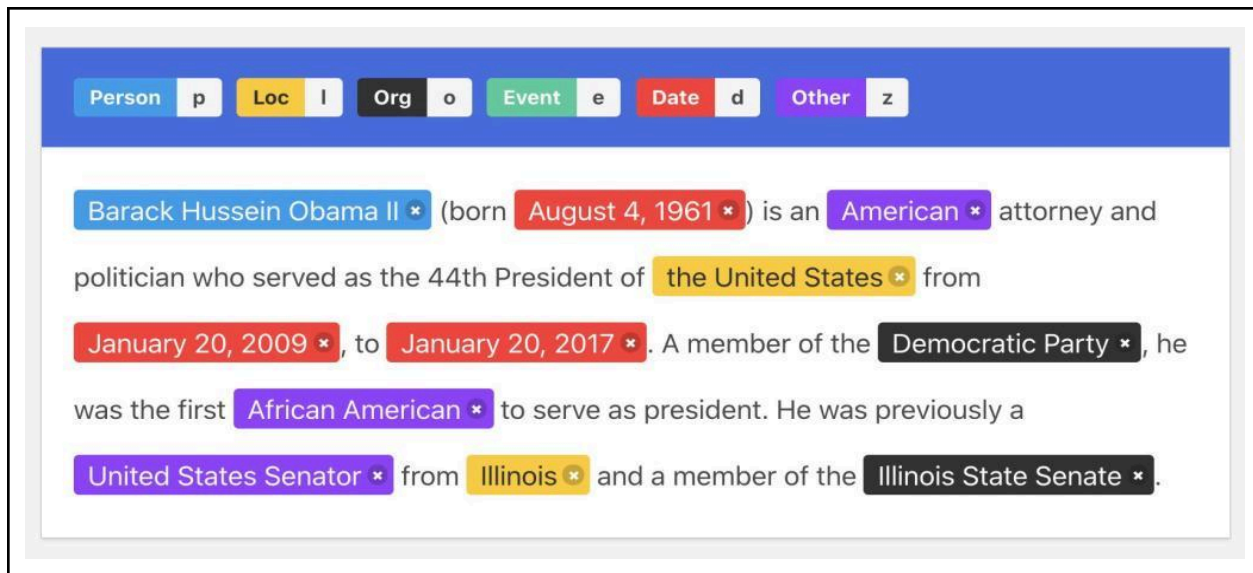
Named entity recognition (NER) : is the IE task of identifying named entities in unstructured data. Named entities are real-world objects that can be uniquely identified. Essentially, they are the proper nouns of data. Named entities include people, dates, corporations, places and products and can be both physical or abstract.

In the sentence “As of January 2025, Arvind Krishna is the CEO of IBM,” the named entities include *January 2025*, *Arvind Krishna*, *CEO* and *IBM*.”

Entity linking:

Entity linking is the process of figuring out whether multiple entities refer to the same real-world object. When conducting IE on an article mentioning “Arvind Krishna,” “Krishna” and “IBM’s CEO,” an entity linking subtask would identify all 3 as references to the same person. Entity linking is also referred to as *coreference resolution*.

Apple ORG today DATE announced the  
second QUANTITY generation iPhone SE COMM  
a powerful new iPhone COMM featuring  
a 4.7- inch QUANTITY Retina HD display.



## Relation extraction (RE)

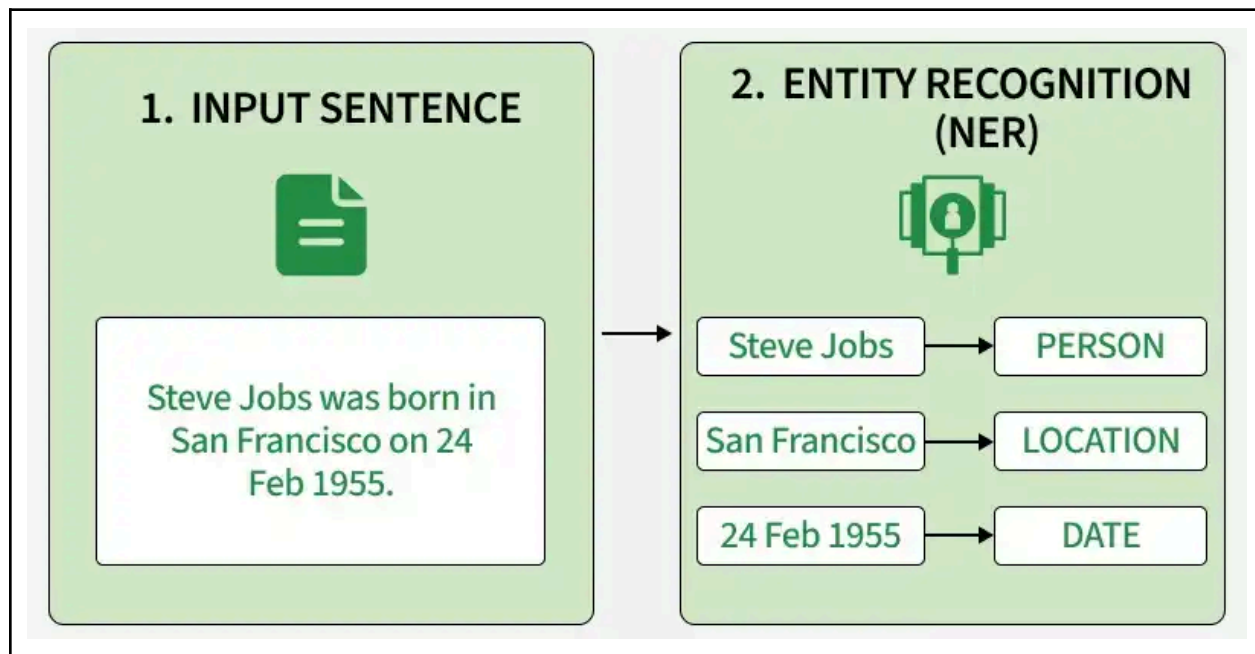
Relation extraction is the information extraction task of identifying and categorizing the relationships between entities in a data source. Uncovering relationships between entities can open the door to insights that might otherwise go unnoticed.

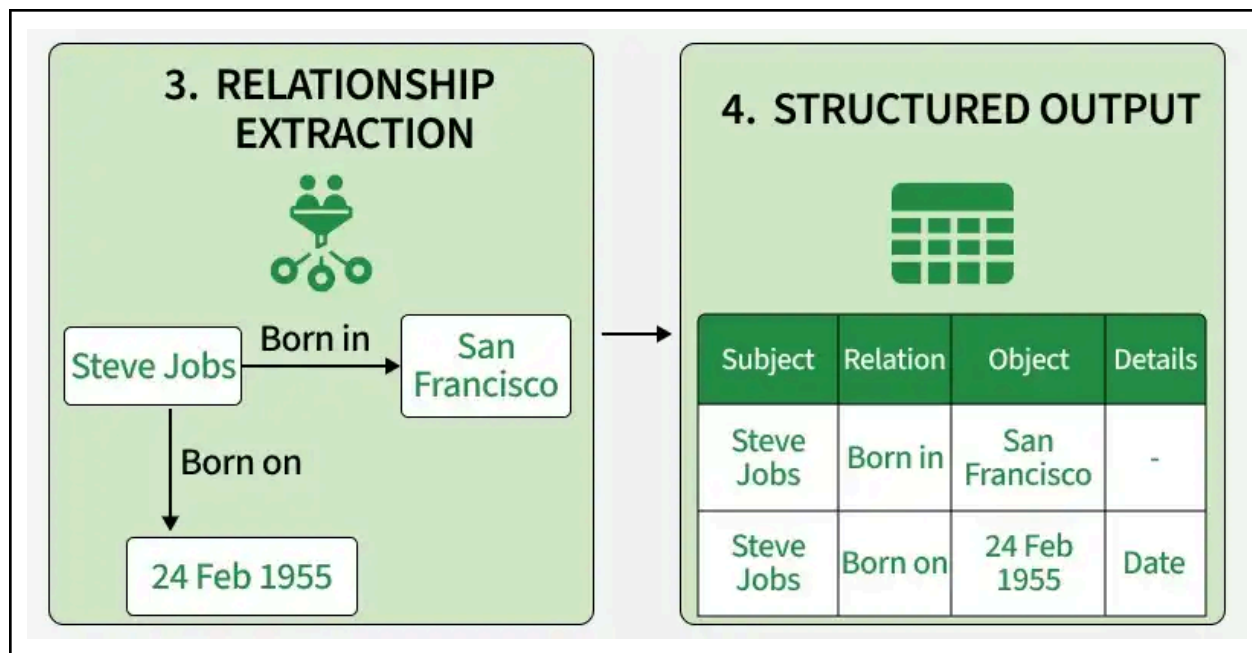
Relationship Extraction (RE) is a Natural Language Processing (NLP) technique used to identify and extract semantic relationships between entities mentioned in textual data. It converts unstructured text into structured information by identifying how different entities are connected.

In our example sentence from the beginning of this section, the RE process would draw a “works at” connection between “Arvind Krishna” and “IBM” with the title of “CEO.”

- Identifies relationships between entities such as people, organizations, locations, and dates.

- Works with Named Entity Recognition (NER) to identify entities before extracting relationships.
- Supports applications such as knowledge graphs, question answering, information retrieval, and recommendation systems.





**Step 1: Input Sentence:** The process begins with an input sentence containing multiple entities and the relationships between them.

**Step 2: Entity Recognition (NER):** The system identifies the named entities present in the sentence:

- Steve Jobs → Person
- San Francisco → Location
- 24 Feb 1955 → Date

**Step 3: Relationship Extraction:** After identifying the entities, the system determines how they are related. In this example, two relationships are extracted:

- Steve Jobs → Born in → San Francisco
- Steve Jobs → Born on → 24 Feb 1955

Components

1. **Entity:** An entity is a person, organization, location, date, product, or any other object mentioned in the text that participates in a relationship.
2. **Entity Pair:** An entity pair refers to the two entities between which a relationship is identified.
3. **Relation:** A relation describes how two entities are semantically connected within a sentence.
4. **Context:** Context refers to the surrounding words or sentence structure that help determine the correct relationship between entities.
5. **Relation Type:** Relation type represents the category of the extracted relationship.

## Event extraction

Event extraction is how IE systems recognize discrete events in a body of input text. Words, such as “appointment” or “meeting,” can trigger an event extraction sequence, as can dates. Event extraction covers the event itself, the time and date at which it occurred and any mentioned participants.

In the sample sentence, “Arvind Krishna attended the conference in January 2025,” an event extraction algorithm would identify that a conference took place in January 2025 and that 1 of the attendees was IBM CEO Arvind Krishna.

Event extraction is the process of gathering knowledge about periodical incidents found in texts, automatically identifying information about what happened and when it happened.

For example:

2018/10 – President Donald Trump’s government banned countries from importing Iranian oil with exemptions to seven countries.

2019/04 – US Secretary of State Mike Pompeo announced that his country would open no more exception after the deadline.

2019/05 – The United States ended with exemptions that allowed countries to import oil from Iran without suffering from US sanctions.

This ability to contextualize information allows us to connect time distributed events and assimilate their effects, and how a set of episodes unfolds through time. Those are valuable insights that drive organizations like [EventRegistry](#) and [Primer.AI](#), which provide the technology to different market sectors.

Sentiment analysis: determines the feeling communicated by a piece of text. Sentiment analysis is a valuable tool for conducting market research and understanding customer behavior.

If given a dataset consisting of user reviews, an IE algorithm can provide semantic insights that reveal the percentages of consumers that feel positively, negatively or neutrally about a product. Product managers could then take those insights and tweak the product to make it more appealing to a greater portion of their current and potential users.

## How does information extraction work?

Information extraction works by parsing unstructured data sources with ML algorithms to identify meaningful data. IE systems label the discovered data entities and store them in an organized, queryable database for efficient retrieval.

Information extraction techniques include:

- Rule-based
- Classification (machine learning)
- Sequence labeling

These methods are not mutually exclusive—advancements in IE have led to hybrid models that combine methods for improved results.

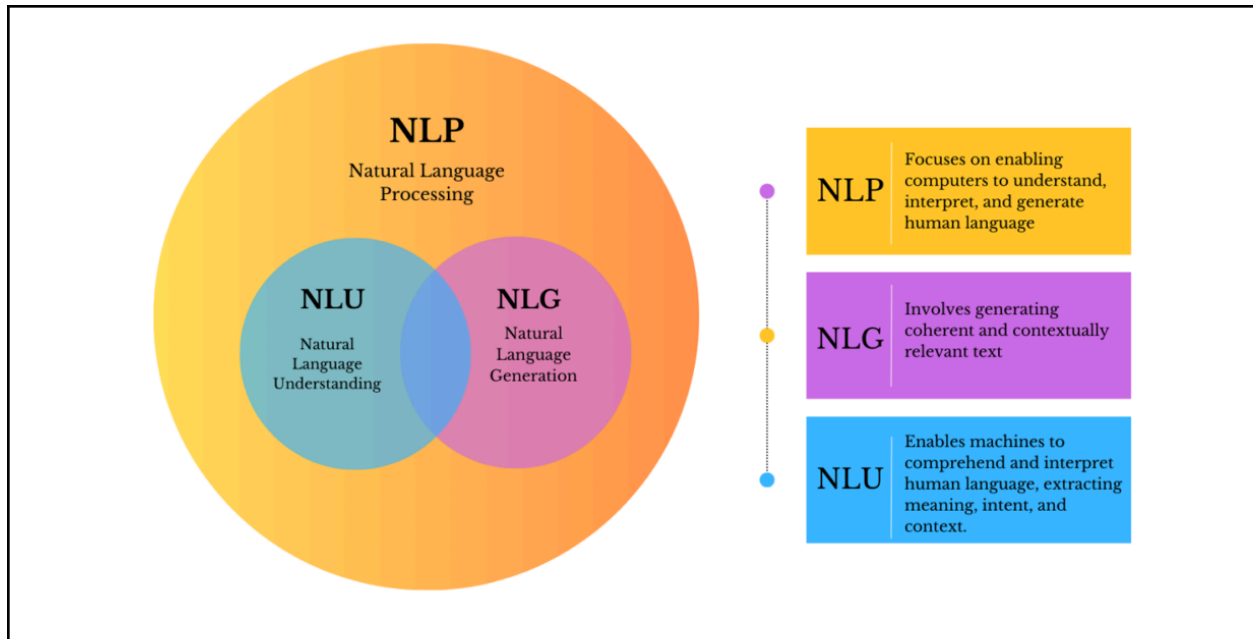
## **NATURAL LANGUAGE PROCESSING:**

Natural language processing (NLP) is a subfield of computer science and artificial intelligence that uses machine learning to enable computers to understand and communicate with human language.

NLP enables computers and digital devices to recognize, understand and generate text and speech by combining computational linguistics (*an interdisciplinary field combining computer science and linguistics to model, analyze, and process human language.*), the rule-based modelling of human language, together with statistical modelling, machine learning and deep learning.

NLP research has helped enable the era of generative AI, from the communication skills of large language models (LLMs) to the ability of image generation models to understand requests.

NLP is already part of everyday life for many, powering search engines, prompting chatbots for customer service with spoken commands, voice-operated GPS systems and question-answering digital assistants on smartphones such as Amazon's Alexa, Apple's Siri and Microsoft's Cortana.



## Benefits of NLP

NLP makes it easier for humans to communicate and collaborate with machines by allowing them to do so in the natural human language they use every day. This offers benefits across many industries and applications.

- Automation of repetitive tasks
- Improved data analysis
- Enhanced search
- Content generation

## Approaches to NLP:

- NLP combines the power of computational linguistics together with machine learning algorithms and deep learning. Computational linguistics uses data science to analyse language and speech.

- It includes two main types of analysis: syntactical analysis and semantical analysis.
- Syntactical analysis determines the meaning of a word, phrase or sentence by parsing the syntax of the words and applying preprogrammed rules of grammar.
- Syntactic analysis (or parsing) focuses on the arrangement of words, phrases, and clauses. It ensures a sentence or statement adheres to the grammatical rules of a language, regardless of whether the sentence makes logical sense.
- Semantical analysis uses the syntactic output to draw meaning from the words and interpret their meaning within the sentence structure.
- Semantic analysis takes the grammatically correct structure and evaluates its actual logical meaning. It analyzes the context of words, the relationship between concepts, and catches logical flaws even if the grammar is perfectly fine.

Three different approaches to NLP include:

**Rules-based NLP** : The earliest NLP applications were simple if-then decision trees, requiring preprogrammed rules. They are only able to provide answers in response to specific prompts. Because there is no machine learning or AI capability in rules-based NLP, this function is highly limited and not scalable.

## **Statistical NLP :**

Developed later, statistical NLP automatically extracts, classifies and labels elements of text and voice data and then assigns a statistical likelihood to each possible meaning of those elements. This relies on machine learning, enabling a sophisticated breakdown of linguistics such as part-of-speech tagging.

## **Deep learning NLP :**

Recently, deep learning models have become the dominant mode of NLP, by using huge volumes of raw, unstructured data both text and voice to become ever more accurate. Deep learning can be viewed as a further evolution of statistical NLP, with the difference that it uses neural network models.

- **Sequence-to-Sequence (seq2seq) models:** Based on recurrent neural networks (RNN), they have mostly been used for machine translation by converting a phrase from one domain (such as the German language) into the phrase of another domain (such as English).
- **Transformer models:** They use tokenization of language (the position of each token words or subwords) and self-attention (capturing dependencies and relationships) to calculate the relation of different language parts to one another. Transformer models can be efficiently trained by using self-supervised learning on massive text databases. A landmark in transformer models was Google's **bidirectional encoder representations from transformers** (BERT), which became and remains the basis of how Google's search engine works.

- **Autoregressive models:** This type of transformer model is trained specifically to predict the next word in a sequence, which represents a huge leap forward in the ability to generate text. Examples of autoregressive LLMs include GPT, Llama, Claude and the open-source Mistral.

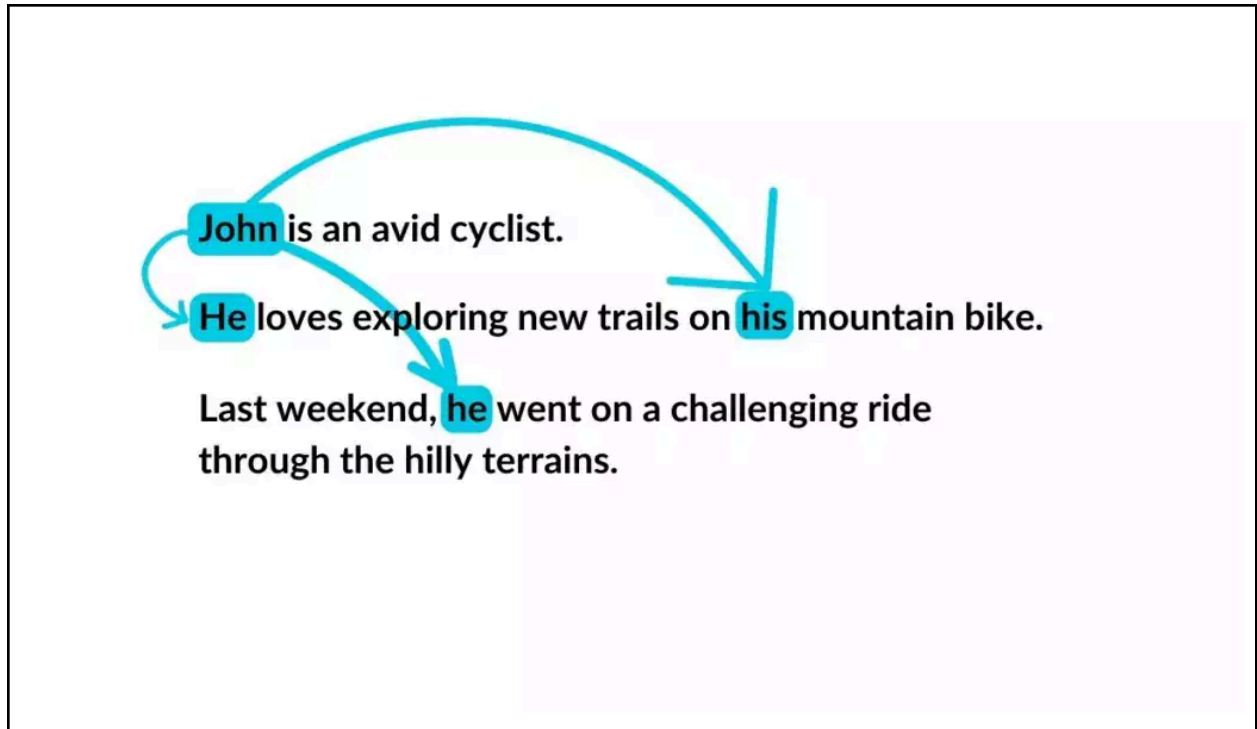
## **NLP Tasks**

Several NLP tasks typically help process human text and voice data in ways that help the computer make sense of what it's ingesting. Some of these tasks include:

- Coreference resolution
- Named entity recognition
- Part-of-speech tagging
- Word sense disambiguation

### **Coreference resolution:**

This is the task of identifying if and when two words refer to the same entity. The most common example is determining the person or object to which a certain pronoun refers (such as “she” = “Mary”). But it can also identify a metaphor or an idiom in the text (such as an instance in which “bear” isn't an animal but a large and hairy person).



### **Named entity recognition (NER):**

NER identifies words or phrases as useful entities. NER identifies “London” as a location or “Maria” as a person's name.

### **Part-of-speech tagging:**

Also called grammatical tagging, this is the process of determining which part of speech a word or piece of text is, based on its use and context. For example, part-of-speech identifies “make” as a verb in “I can make a paper plane,” and as a noun in “What make of car do you own?”

### **Word sense disambiguation:**

This is the selection of a word meaning for a word with multiple possible meanings. This uses a process of semantic analysis to examine the word in context. For example, word sense disambiguation helps distinguish the meaning of the verb “make” in “make the grade” (to achieve) versus “make a bet” (to place). Sorting out “I will be merry when I marry Mary” requires a sophisticated NLP system.

### **How NLP works ?**

NLP works by combining various computational techniques to analyze, understand and generate human language in a way that machines can process. Here is an overview of a typical NLP pipeline and its steps:

### **Text preprocessing :**

NLP text preprocessing prepares raw text for analysis by transforming it into a format that machines can more easily understand. It begins with tokenization, which involves splitting the text into smaller units like words, sentences or phrases. This helps break down complex text into manageable parts. Next, lowercasing is applied to standardize the text by converting all characters to lowercase, ensuring that words like "Apple" and "apple" are treated the same. Stop word removal is another common step, where frequently used words like "is" or "the" are filtered out because they don't add significant meaning to the text. Stemming or lemmatization reduces words to their root form (e.g., "running" becomes "run"), making it easier to analyze

language by grouping different forms of the same word. Additionally, text cleaning removes unwanted elements such as punctuation, special characters and numbers that may clutter the analysis.

After preprocessing, the text is clean, standardized and ready for machine learning models to interpret effectively.

### **Feature extraction**

Feature extraction is the process of converting raw text into numerical representations that machines can analyze and interpret. This involves transforming text into structured data by using NLP techniques like Bag of Words and TF-IDF, which quantify the presence and importance of words in a document. More advanced methods include word embeddings like Word2Vec or GloVe, which represent words as dense vectors in a continuous space, capturing semantic relationships between words. Contextual embeddings further enhance this by considering the context in which words appear, allowing for richer, more nuanced representations.

### **Text analysis**

Text analysis involves interpreting and extracting meaningful information from text data through various computational techniques. This process includes tasks such as part-of-speech (POS) tagging, which identifies grammatical roles of words and named entity recognition (NER), which detects specific entities like names, locations and dates.

## **Model training :**

Processed data is then used to train machine learning models, which learn patterns and relationships within the data. During training, the model adjusts its parameters to minimize errors and improve its performance. Once trained, the model can be used to make predictions or generate outputs on new, unseen data. The effectiveness of NLP modeling is continually refined through evaluation, validation and fine-tuning to enhance accuracy and relevance in real-world applications.

NLP use cases by industry: NLP applications can now be found across virtually every industry.

### **Finance:**

In financial dealings, nanoseconds might make the difference between success and failure when accessing data, or making trades or deals. NLP can speed the mining of information from financial statements, annual and regulatory reports, news releases or even social media.

### **Healthcare:**

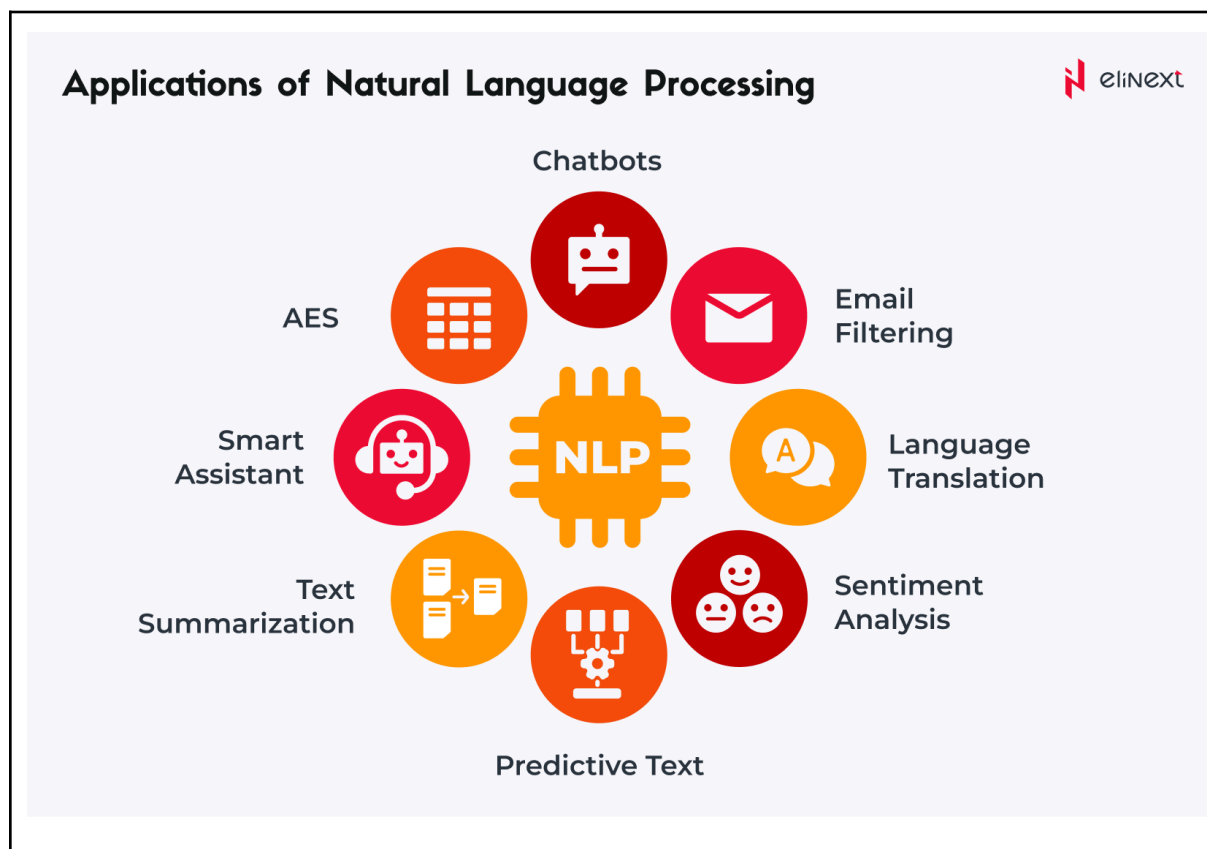
New medical insights and breakthroughs can arrive faster than many healthcare professionals can keep up. NLP and AI-based tools can help speed the analysis of health records and medical research papers, making better-informed medical decisions possible, or assisting in the detection or even prevention of medical conditions.

### **Insurance:**

NLP can analyze claims to look for patterns that can identify areas of concern and find inefficiencies in claims processing, leading to greater optimization of processing and employee efforts.

**Legal :**

Almost any legal case might require reviewing mounds of paperwork, background information and legal precedent. NLP can help automate legal discovery, assisting in the organization of information, speeding review and making sure that all relevant details are captured for consideration.



## **Understanding the Basics of Content Analysis:**

So, how do you get a grip on what people are actually doing, thinking, or feeling? The answer lies in content analysis; a research technique that helps you decode the complexities of consumer behaviour, without having to guess your way through it.

Content analysis isn't just about reading between the lines. It's about digging deeper, uncovering hidden patterns, and making sense of textual data. Whether it's social media posts, customer feedback, or interviews, this method helps you transform raw text into valuable insights.

### **What is Content Analysis?**

Content analysis is a research method used to interpret textual data. It's not just about counting words or spotting recurring themes; it's about understanding the context and the underlying message.

The goal here is to break down complex data into manageable bits and find patterns that tell a story. You can use it to analyse anything from social media comments to long-form interviews, making it a versatile tool for anyone in research or marketing.

## Quantitative vs. Qualitative Content Analysis

There are two main flavours of content analysis:

- Quantitative: Focuses on counting and measuring. For example, how many times does a certain keyword appear?
- Qualitative: Seeks to understand the deeper meaning behind the data. Why are people saying what they're saying?

Both approaches have their place, and the trick is knowing when to count and when to interpret.

## Content Analysis vs. Thematic Analysis

If you've ever confused content analysis with thematic analysis, you're not alone. They do share similarities, but the difference lies in their structure and purpose.

### Content Analysis

- More structured and systematic
- Coding categories are established beforehand
- Aimed at quantifying data

## Thematic Analysis

- Flexible and interpretive
- Themes emerge through the analysis process
- Focused on understanding patterns of meaning

In short, content analysis is like having a map before you start, while thematic analysis is more of a treasure hunt where you discover the map along the way.

## The Mechanics of Content Analysis

Getting started with content analysis isn't just about diving into the data. It's about planning, coding, and interpreting, all while keeping your research question in focus.

### **Step 1: Define Your Research Question**

Everything starts with a clear question. Are you trying to understand customer sentiment about a new product? Or perhaps you're exploring how users describe their pain points?

### **Step 2: Collect Your Data**

Gather the texts you'll be analysing. This could be social media comments, customer reviews, interview transcripts, or even emails.

### **Step 3: Coding**

Coding is where you break the content into manageable chunks and assign labels to these chunks. This could be a word, phrase, or even a recurring idea.

### **Step 4: Categorisation**

Next, group similar codes together into categories. This helps to see the bigger picture and makes it easier to spot patterns.

### **Step 5: Interpretation**

Now comes the fun part, making sense of it all. Analyse your categories, look for themes, and draw conclusions that align with your research question.

## **Spotting Patterns: The Heart of Content Analysis**

Patterns are where the magic happens. You're not just looking for what's repeated but also for what's missing. It's not just about what people say but how they say it.

## **Real-Life Applications of Content Analysis**

Content analysis isn't just for academic papers; it's a powerhouse for real-world insights.

### **Marketing**

Brands use content analysis to understand customer sentiment, track campaign performance, and spot emerging trends. Analysing customer feedback helps marketers craft more targeted messages.

### **UX Design**

Ever wondered why users abandon your app halfway through? Analysing in-app feedback or support tickets can reveal usability issues you didn't know existed.

## **Social Media Monitoring**

Want to know what people really think about your brand?

Analysing tweets, comments, and reviews gives you a pulse on public opinion; warts and all.

## **The AI Factor: Automating Content Analysis**

Artificial intelligence is transforming content analysis by making coding faster and more consistent. AI can process huge volumes of data and even detect sentiment, making it a game-changer for marketers and researchers alike. But remember, AI doesn't replace human judgement. It's great for spotting patterns, but the real insight comes from interpreting what those patterns mean.

## **Navigating Challenges and Ethical Concerns**

Content analysis has its quirks. You'll need to watch out for:

- **Bias:** Your own perceptions can influence coding.
- **Privacy Issues:** Always get consent when analysing personal data.
- **Data Overload:** Focus on relevant data to avoid getting overwhelmed.

The key is to approach content analysis methodically, keeping your research question front and centre.

## **Wrapping Up: Why Content Analysis Matters**

Content analysis is more than just a research method; it's a window into your users' world. It helps you understand what drives them, what frustrates them, and how you can better meet their needs.

Whether you're tweaking a product, crafting a marketing campaign, or just trying to figure out what people are saying about your brand, content analysis gives you the insights to make informed decisions.

# Sentiment Analysis Using Natural Language Processing (NLP).

- Sentiment analysis, also known as opinion mining, is a technique used in natural language processing (NLP) to identify and extract sentiments or opinions expressed in text data.
- The primary objective of sentiment analysis is to comprehend the sentiment enclosed within a text, whether positive, negative, or neutral.
  1. How does sentiment analysis work ?
  2. Everyday use cases for sentiment analysis.
  3. Challenges in sentiment analysis.
  4. Description of Natural Language Processing (NLP) techniques
  5. Developing sentiment analysis machine learning model

## How sentiment analysis works:

- Sentiment analysis is a process that involves analyzing textual data such as social media posts, product reviews, customer feedback, news articles, or any other form of text to classify the sentiment expressed in the text.

- The sentiment can be classified into three categories: Positive Sentiment Expressions indicate a favorable opinion or satisfaction; Negative Sentiment Expressions indicate dissatisfaction, criticism, or negative views; and Neutral Sentiment Text expresses no particular sentiment or is unclear.
- Before analyzing the text, some preprocessing steps usually need to be performed. These include **tokenization**, breaking the text into smaller units like words or phrases, **removing stop words** such as common words like “and,” “the,” and so on, and **stemming or lemmatization**, which involves reducing words to their base or root form. At a minimum, the data must be cleaned to ensure the tokens are usable and trustworthy.

The strategies vary in complexity as well. In order of complexity

**1. Lexicon-based sentiment analysis** is a technique used in natural language processing to detect the sentiment of a piece of text. It uses lists of words and phrases (lexicons or dictionaries) that are linked to different emotions to label the words (e.g. positive, negative, or neutral) and detect sentiment.

In our example, the words are labeled with the help of a so-called valence dictionary. Each word in the text can have some type of emotional valence, such as “great” (positive valence) or “terrible” (negative valence), which gives us a positive or negative impression of the airline.

Take the phrase “Good airlines sometimes have bad days.”. A valence dictionary would label the word “Good” as positive; the word “bad” as negative; and possibly the other words as neutral.

Once each word in the text is labeled, we calculate an overall sentiment score by counting the numbers of positive and negative words, and then combining the values.

A popular formula to calculate the sentiment score (StSc) is:

$$\text{StSc} = \frac{\text{number\_of\_positive\_words} - \text{number\_of\_negative\_words}}{\text{total\_number\_of\_words}}$$

If the sentiment score is negative, the text is classified as negative. A positive score means a positive text, and a score of zero means the text is classified as neutral.

In the lexicon-based approach we can skip the whole process of building a machine learning model. Instead, we can analyze the overall sentiment on-the-fly, calculating sentiment scores based on our valence dictionary.

**Machine Learning and Deep Learning:** One approach to classify sentiments is to use supervised learning algorithms or neural networks. These methods rely on pre-labelled data to accurately categorise different emotions or opinions.

This method involves training a model to identify the sentiment of a piece of text. It's based on a set of labeled training data. In this case, we first of all give our model data on what positive and negative text will look like. We call this our training data. Our model would use training data to predict the positivity scores of other pieces of text.

Machine Learning can be more accurate than Lexicon-based analysis. However, it requires large amounts of labelled training data to achieve that amount of accuracy. Therefore, it can be more resource-heavy.

# **Pre-trained transformer-based deep learning**

Deep Learning and Neural Networks are the foundational blocks of technology such as GPT-4 and BERT.

This approach involves using technologies trained on massive amounts of data. They can understand the meaning of the text and more accurately deduce if a piece of text is positive or negative.

This approach produces highly accurate data. However, it may not be the best approach in some cases. It is computationally expensive and best suited for serious sentiment analysis.

**Hybrid Approaches:** Combining multiple methods to improve accuracy, like machine learning models and lexicon-based analysis.

## **Challenges in sentiment analysis**

It can be challenging for computers to understand human language completely. They struggle with interpreting sarcasm, idiomatic expressions, and implied sentiments. Despite these challenges,

sentiment analysis is continually progressing with more advanced algorithms and models that can better capture the complexities of human sentiment in written text.

### **Challenge 1: Sarcasm**

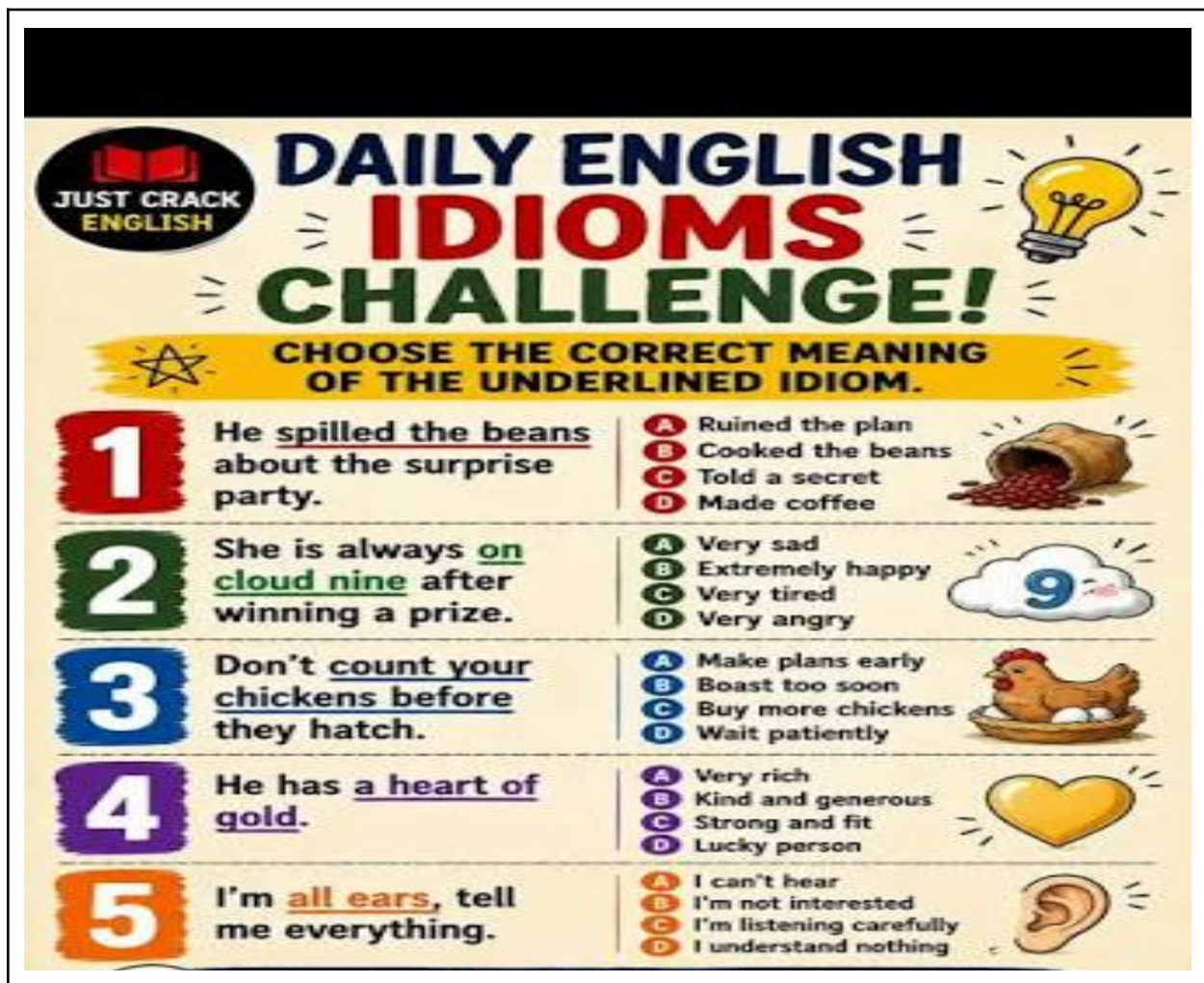
People use sarcasm and irony in casual conversations on social media. This act of expressing negative sentiment using backhanded compliments allows sarcasm to easily cheat sentiment analysis models unless they're specifically designed to take its possibility into account.

**solution-** A good sentiment analysis API like **Bytesview** or any other API will be able to detect the context of the language used in creating actual sentiment when something is posted. Bytesview is trained for 30+ language datasets on which the sentiment analysis model works which gives precise and accurate results.

### **Challenge 2: Idioms**

Machine learning programs don't understand a figure of speech usually. Idioms boggle the algorithm because it understands things in the literal sense. the sentence can be misconstrued by the algorithm or even ignored when an idiom is being used in a comment or review. The sentiment analysis platform needs to be trained in understanding idioms to overcome this problem. The problem becomes manifold when it comes to multiple languages.

**Solution-** The only way this challenge can be solved with sentiment analysis accuracy is if the neural networks of the API are trained enough to understand and interpret idioms. Idioms are mapped according to nouns that denote emotions like joy, anger, success, determination, etc, and then the models are trained accordingly, only then can a tool for analyzing sentiment give accurate insights from such text.



First, let's import all the Python libraries that we will use throughout the program.

## Basic Python Libraries

1. **Pandas** — For data analysis and data manipulation.
2. **Matplotlib** — For data visualization.
3. **Seaborn** — It's based on matplotlib and provides a high-level interface for data visualization.
4. **WordCloud** — For visualizing text data in the form of clouds.

5. **re** — It provides functions to pre-process the strings as per the given regular expression.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
import re
```

## Natural Language Processing

1. **nltk** — Natural Language Toolkit is a collection of libraries for natural language processing.
2. **stopwords** — A collection of words that don't provide any meaning to a sentence.
3. **WordNetLemmatizer** — It is used to convert different forms of words into a single item but still keeping the context intact.

```
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
```

## Scikit-Learn (Machine Learning Library for Python)

1. **CountVectorizer** — For transforming text into vectors.
2. **GridSearchCV** — For hyperparameter tuning of ML models.
3. **RandomForestClassifier** — Machine learning algorithm for classification.

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import GridSearchCV
from sklearn.ensemble import RandomForestClassifier
```

## Evaluation Metrics

1. **Accuracy Score** — It is no. of correctly classified instances/total no. of instances.
2. **Precision Score** — It is the ratio of correctly predicted instances over total positive instances.
3. **Recall Score** — It is the ratio of correctly predicted instances over total instances in that class.
4. **Roc Curve** — A plot of true positive rate against false positive rate.
5. **Classification Report** — Report of precision, recall and f1 score.
6. **Confusion Matrix** — A table used to describe the classification models.

```
from sklearn.metrics import  
accuracy_score, precision_score, recall_score, confusion_matrix, ro  
c_curve, classification_report  
from scikitplot.metrics import plot_confusion_matrix
```

## Preprocessing

To prepare our data for model training, we need to convert our text data into features that our model will use to train and cast future predictions. We'll use two preprocessing steps:

- Count vectorizing text
- tf-idf

Count vectorisation is a technique in NLP that converts text documents into a matrix of token counts. Tokens can be words, characters, or n-grams. Each token represents a column in the matrix, and the resulting vector for each document has counts for each token.

Here's an example of how we transform the text into features for our model. The corpus of words represents the collection of text in raw form we collected to train our model[3].

```
corpus = [
    'This is the first document',
    'This document is the second document',
    'and this is the third one',
    'is this the first document'
]

vec = CountVectorizer().fit(corpus)
vec.get_feature_names()
>>>
['and,' 'document,' 'first,' 'is,' 'one,' 'second,' 'the,' 'third,' 'this']

vec.transform(corpus).toarray()
>>>
array([[0, 1, 1, 1, 0, 0, 1, 0, 1],
       [0, 2, 0, 1, 0, 1, 1, 0, 1],
       [1, 0, 0, 1, 1, 0, 1, 1, 1],
       [0, 1, 1, 1, 0, 0, 1, 0, 1]])
```

The purpose of using tf-idf instead of simply counting the frequency of a token in a document is to reduce the influence of tokens that appear very frequently in a given collection of documents.

Here's an example of our corpus transformed using the tf-idf preprocessor[3]

```
vectorized = vec.transform(corpus).toarray()
tfidf = TfidfTransformer().fit(vectorized)

tfidf.transform(corpus).toarray()
>>>
array([[0.00, 0.46, 0.58, 0.38, 0.00, 0.00, 0.38, 0.00, 0.38],
       [0.00, 0.68, 0.00, 0.28, 0.00, 0.53, 0.28, 0.00, 0.28],
       [0.51, 0.00, 0.00, 0.26, 0.51, 0.00, 0.26, 0.51, 0.26],
       [0.00, 0.46, 0.58, 0.38, 0.00, 0.00, 0.38, 0.00, 0.38]])
```

## CASE STUDY 1 -Developing sentiment analysis machine learning model:

<https://medium.com/analytics-vidhya/nlp-getting-started-with-sentiment-analysis-126fcd61cc4a>

**Web analytics** is the gathering, synthesizing, and analysis of website data with the goal of improving the website user experience.

- It's a practice that's useful for managing and optimizing websites, web applications, or other web products.

- It's a data-driven practice that helps companies make high-impact website decisions.
- You might also get ideas on improving your website and driving business growth from web analytics.
- Product managers, data scientists, UX designers, and others can use web analytics to enhance their website or product experience and meet customer needs

Web analytics refers to the process of collecting website data and then processing, reporting, and analyzing it to create an online strategy for improving the website experience

Web analytics is important to help you:

- Refine your marketing campaigns
- Understand your website visitors
- Analyse website conversions
- Improve the website user experience

- Boost your search engine rankings
- Choose the right referral sources
- Track digital marketing efforts
- Boost online sales
- Use a web analytics tool to automatically track key site performance metrics, like bounce rate, conversion rate, monthly unique visitors, and more.
- Some common issues with web analytics are that it can be overwhelming to keep track of so many metrics, data is not always accurate, and data privacy could be at risk.
- Find an analytics tool that addresses these concerns effectively.

## Bounce Rate

- **Definition:** The percentage of visitors who land on your website and leave right away without clicking to another page or doing anything else.
- **What it shows:** If people are confused, or if the page does not match what they wanted.
- **Goal:** A lower bounce rate is usually better, meaning people stay and look around

## Conversion Rate

- **Definition:** The percentage of visitors who complete a goal you want them to do, like buying a product or signing up for an email.
- **What it shows:** How well your website turns casual visitors into paying customers or leads.
- **Goal:** A higher conversion rate is better, meaning your site successfully drives action

## What is web analytics used for?

Web analytics is helpful for understanding which channels bring users to your website. You can also identify popular site content by calculating the average stay on your web pages and how users interact with them—including which pages prompt users to leave. Web analytics features like Amplitude's Journeys show how users interact with your site.

You can use this information to improve underperforming pages and increase audience engagement across your website. For example, payment processor Square used web analytics from Amplitude to redesign its homepage, creating a new seller journey and making it easier for interested companies to get started with the company. The company continued to iterate on its redesign, gathering data and saw double-digit improvements in key metrics.

One key insight: Customers preferred chatting with company representatives over submitting their information in a contact form.

## Categories of web analytics

The data you gather about your company's online performance may come from multiple sources. Two main web analytics types are defined by the source of the data.

### On-site web analytics

On-site web analytics measure your audience's behaviors and interactions directly with your website. This category of analytics can help you track engagement and conversions. Amplitude offers true self-serve analytics to help you understand everything that happens on your site.

**Amplitude** is a powerful digital analytics platform that tracks user behavior on websites and apps. Instead of just counting page views, it focuses on event-based data like clicks, sign-ups, and purchases to

show how people move through your site and why they convert or leave.

The most common type of on-site analytics is page tagging. When a page is "tagged" with a snippet of code, it collects information from each visitor. Page tagging enables you to follow visitors as they engage with your site and watch their path to bounce or conversion. Because you get detailed and real-time data, on-site analytics help you learn what parts of your site are most effective versus those that lose readers' interest before winning their conversions.

## **Off-site web analytics**

Off-site analytics brings in third-party data suppliers to help you understand your site's performance as part of the overall market. This category encompasses everything from social media analytics to panel data. These analytics aren't just about your site: They're about how you stack up to your competitors.

For instance, your social media analytics tool may tell you your share of social voice, but also that of two major companies also operating in your niche. Your on-site analytics tools might show a

steady stream of new visitors from social media, which makes your Instagram and TikTok efforts look great. However, imagine knowing that your company only comes up in 10% of relevant discussions, whereas a competitor's name gets dropped in nearly 50% of those conversations. Clearly, you have a greater potential audience on these sites and plenty of work to do if you want to overtake your competitor.

## Understand your website visitors

Web analytics tools reveal key details about your site visitors—including their average time spent on page and whether they're a new or returning user—and which content draws in the most traffic. With this information, you'll learn more about what parts of your website and product interest users and potential customers the most.

For instance, an analytics tool might show you that most of your visitors land on your German site. You could use this information to ensure that your product has a German version that's well-translated to meet the needs of these users.

# Analyze website conversions

Conversions could mean real purchases, signing up for your newsletter, or filling out a contact form on your website. Web analytics can give you information about the total number of these conversions, how much you earned from the conversions, the percentage of conversions (the number of conversions divided by the number of website sessions), and the abandonment rate. You can also see users' conversion paths, showing how your users moved through your site before they converted.

## Boost your search engine optimization (SEO)

Connect your web analytics tool with [Google Search Console](#) to track which search queries are generating the most traffic for your site. With this data, you'll know what topics to focus on as you try to boost your site's rankings.

It's also possible to set up on-site search tracking to know what users search for on your site. This search data can help you generate content ideas for your site, especially if you have a blog.

## Understand top-performing content

[Web analytics tools](#) also help you learn what site content is performing the best so you can focus on the topics and formats your audience likes. You can also use this information to make product improvements.

For instance, you may notice blog articles that talk about design are the most popular on your website. This might signal that your users care about your product's design features, leading you to invest more resources there. Popular content on your website could spark ideas for new product features, too.

## Understand and focus on strong referral sources

Web analytics tell you who your top referral sources are, so you know which channels to focus on. If you're getting 80% of your traffic from Instagram, your company's marketers will know to invest in ads on that platform.

Web analytics also show you which outbound links people are clicking on. Your company's marketing team might discover a mutually beneficial relationship with these sites or brands and reach out to them to explore partnership or cross-referral opportunities.

# Measure the success of digital marketing campaigns with web analytics

Effective digital marketing campaigns respond to audience insights. Web analytics can help you [track viewers' responses](#) to your marketing efforts and tailor them to fit the demographics of your target audience. You want to know what users are responding to and why.

Digital marketing campaigns rely on a bevy of analytics to learn these answers. The metrics you use will depend on the type of campaign you're running and the channel you're using to run it. Some of the most common [digital marketing web metrics](#) are:

- [Clickthrough rate \(CTR\)](#): How many ad views result in reader clicks.
- [Conversion rate](#): The percentage of viewers who ultimately complete the desired action.
- Bounce rate: The percentage of people who, after clicking on your ad, leave your site after seeing only the landing page they were taken to.
- Engagement time/time on page: How long the average individual spends browsing the landing page your ad sent them to.

Beyond these simple metrics, digital marketing teams may track respondents' journeys across their site. For ecommerce-focused campaigns, additional metrics like customer acquisition cost and average order value help show the ROI of a digital marketing campaign.

## The web analytics process and eight metrics to track

### The web analytics process

Website performance metrics vary from company to company based on their goals for their site. However, successful web analytics programs follow the same process:

- Set business goals: Define the key metrics that will determine the success of your business and website.
- Collect data: Gather information, statistics, and data on website visitors using analytics tools.
- Process data: Convert the raw data you've gathered into meaningful ratios, KPIs, and other information that tells a story.
- Report data: Display and share the processed data in an easy-to-read format.

- Develop an online strategy: Build a website experience that supports your business goals.
- Experiment: Run [A/B tests](#) to determine what your audience responds to, and continue working to improve your reader experience.

Web analytics tools can collect countless pieces of data, even from the simplest website.

### **Clickstream Analysis**

- **Clickstream analysis** is the process of analysing user behaviour on websites, mobile apps, and other digital platforms.
- This analysis involves collecting data on user interactions, such as clicks, page views, and other actions, and using that data to gain insights into user behaviour patterns.

(or)

Clickstream data and clickstream analytics are the processes involved in collecting, analyzing and reporting aggregate data about which pages a website **visitor visits—and in what order**. The path the visitor takes through a website is called the clickstream.

- Clickstreams are categorised into **clickstream data** and **clickstream analytics**, which is also referred to as clickstream analysis.
- The clickstream data is the information collected about a user while they browse through a website or use a web browser.
- Clickstream analytics is the process of tracking, analyzing and reporting data on the pages a user visits and user behavior while on a webpage.
- Websites use clickstream data to show how a user progressed from an initial search or landing page to buying an item or service.
- Search engines use clickstream data sets to show where a user has searched for a term, when they have clicked on it and if they go back to searching after this.
- Internet service providers, advertising networks, and IT and telecom organizations also collect clickstream data.

### **Clickstream data metrics**

Clickstream data includes the following types of web analytics information:

- whether the individual is a unique or repeat visitor to the website
- the terms an individual plugs into a search engine;

- what page the individual lands on first;
- the amount of time a user spends on a page;
- the features on the page the user clicks on and engages with
- when and where an item is added or removed from a cart
- where the user goes next; and
- when the back button is used.

Clickstream data collected from a single session of a user interacting with a website may not be useful. However, an organization can use aggregate data gathered from many visitors to improve its website or service.

For example, if a lot of visitors leave a site after landing on a page with too little information, the organization may need to enhance the page with more valuable information. Likewise, if visitors often land on a page that isn't the website's homepage, then the organization may want to redesign that page to be more inviting and informative to users.

Clickstream data does not include personal details about a user, and it is typically stored on the server that supports the website

## **Why is Clickstream Analysis Important?**

Clickstream analysis is important for businesses and organizations because it provides insights into how users interact with their digital properties. By understanding how users behave, businesses can optimize their digital experiences, improve conversion rates, and increase revenue.

For example, clickstream analysis can help businesses identify which pages on their website are most popular, how long users spend on each [page](#), and which buttons or links users are clicking. This information can then be used to optimise the website layout, improve navigation, and make it easier for users to find what they are looking for.

## Clickstream analytics and how it is used

Organisations use clickstream analytics to uncover trends and draw conclusions from different metrics about their websites. This process typically uses a web server log file to monitor user activity on a website.

Using the clickstream analysis, an organization can collect data on the number of page visits, views, and unique and repeat visitors. This data provides an idea of how the organization's website performs, and it can help approximate the typical user experience (UX). A website owner can then adjust the site to make it more

user-friendly and increase the chance that visitors will stay longer. make a purchase or otherwise interact with the website and the organization behind it.

Because an extremely large volume of data can be gathered through clickstream analysis, many e-businesses rely on big data analytics and related tools such as Hadoop to interpret the data and generate reports on specific areas of interest.

There are two levels of clickstream analysis: **traffic analytics** and **e-commerce analytics**.

### **Traffic analytics**

This analysis operates at the server level. It collects and analyzes the following data sets:

- how many pages are served to a user;
- how long it takes each page to load;
- how often the user hits the browser's back button; and
- how much data is transmitted before the user moves to a different webpage.

## **E-commerce analytics**

This analysis uses clickstream data to determine the effectiveness of a website in terms of conversions and transactions. It is concerned with the following data points:

- what pages the shopper lingers on;
- what the shopper puts in or takes out of a shopping cart;
- what items the shopper purchases;
- whether the shopper belongs to a loyalty program;
- whether the shopper uses a coupon code; and
- the shopper's preferred method of payment.

## **Benefits of clickstream data analysis**

There are a number of benefits organizations can get from clickstream data and clickstream analytics. Among them are the following:

- **User information:** The data collected can include search terms used, pages landed on, webpage features used and the addition or removal of items from a cart, all of which can lead to more actionable insights.

- **User routes:** Organizations can use data analysis to view the different routes their online visitors or customers take to reach a page or to make a purchase.
- **Customer trends and insights:** Collecting and analyzing the clickstreams of a large number of visitors lets an organisation identify trends in the following areas:
  - how visitors get to the website;
  - what they do once there;
  - how long they stay on a page;
  - the number of page visits visitors make; and
  - the number of unique and repeat visitors.
- **UX:** If a majority of users quickly leave a page or website, it could be a sign that the page is poorly optimized or doesn't contain enough information of value. Clickstream data enables an organization to recognize UX shortcomings, enabling them to make necessary changes.
- **Digital marketing:** Clickstream data can be used to determine the amount of traffic coming from ad banners and campaigns. Such data provides insight as to which advertisements are most effective and lead to customer conversion rate optimization. Clickstream analysis can also

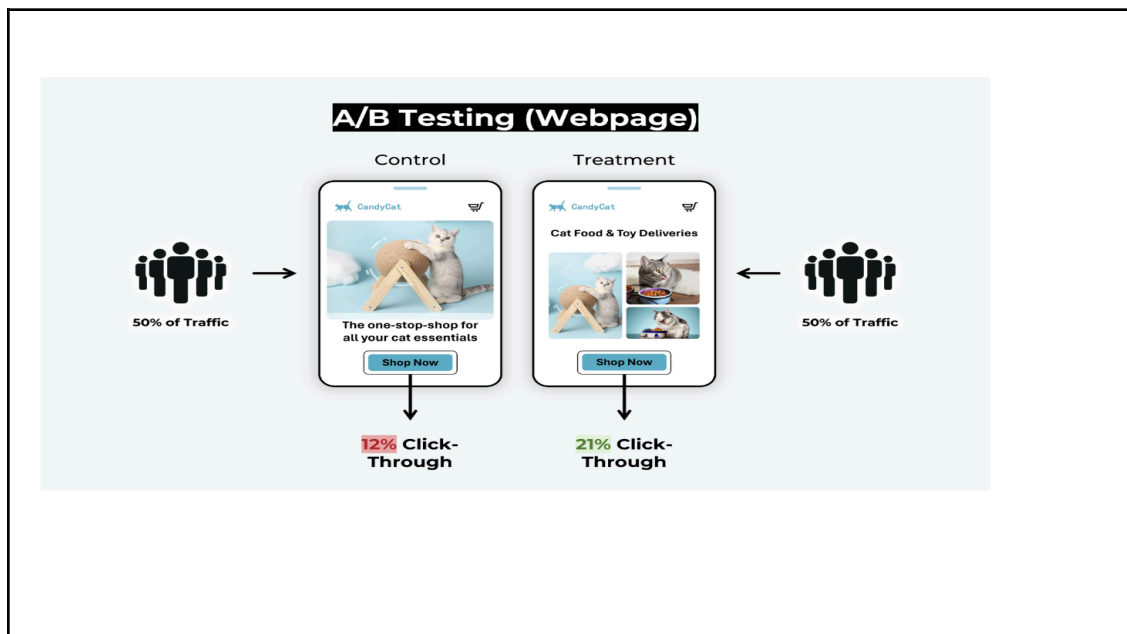
derive what times of day, month or year a marketing strategy is most effective.

## **How does Clickstream analysis work ??**

- Clickstream analysis works by collecting and analyzing data on user interactions with digital properties.
- This data is typically collected using tracking codes or cookies, which allow businesses to track user behavior over time and across multiple sessions.
- Once the data is collected, it is analyzed using data analytics tools and techniques to identify patterns and trends in user behavior.
- This analysis can include visualizations, such as heat maps or funnels, to help businesses understand how users are interacting with their digital properties.
- In addition to analyzing user behavior on their own websites and apps, businesses can also use clickstream analysis to gain insights into user behavior on third-party sites and platforms.
- For example, businesses can analyze user behavior on social media platforms or eCommerce marketplaces to identify new opportunities for engagement and revenue generation.

# What is A/B testing?

An A/B test is an experiment that compares two or more versions of something (like a webpage, advertisement, or message) to figure out which one performs better. That generally involves splitting people into groups who each see a different version and then measuring which version produces the most clicks or customers.



One of the biggest challenges with running an A/B test is that you need a LOT of people to participate in order to generate statistically

valid results. Ronny Kohavi (former Vice President of Airbnb, Microsoft and Amazon) says that “unless you have at least tens of thousands of users for an A/B test, the statistics just don't work out. A retail site trying to detect changes will need something like 200,000 users”.

Well, that's disappointing... Most of us don't have 200,000 people waiting around for an experiment! The good news is, there are plenty of ways you can use the principles of A/B testing with much smaller sample sizes when comparing things like design mockups, concept images, ad copy, problem statements, sales messaging, and loads more. So what would a survey-based A/B test look like?

## **How to run A/B tests using online surveys**

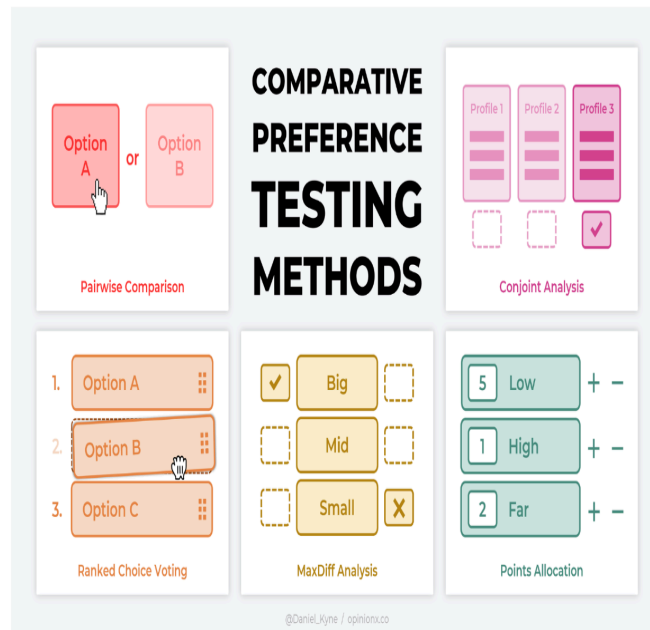
There are two ways you can conduct A/B tests using online surveys:

# 1. Comparative Preference Testing

Comparative Preference Testing (also known as discrete choice analysis) is a research technique that forces people to compare options head-to-head and analyzes their votes to calculate which options they like or dislike most.

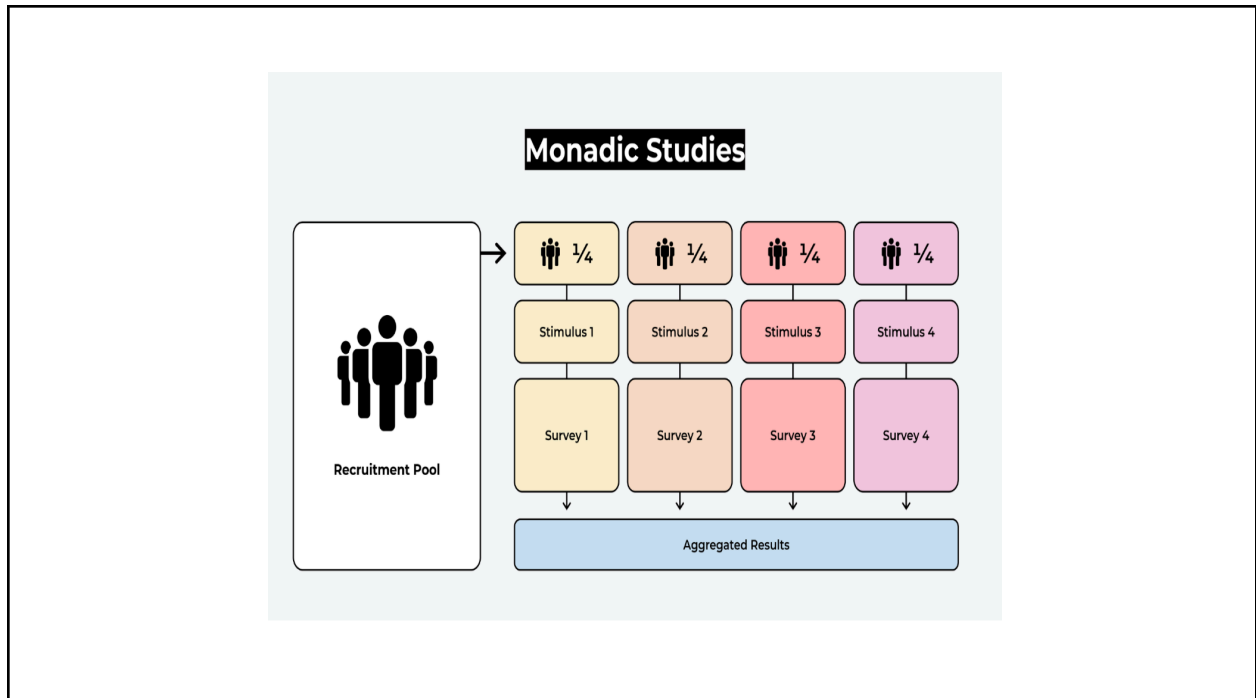
Question formats that use forced comparison and are suitable for A/B testing surveys include:

- Pairwise Comparison — shows two options at a time in a series of head-to-head votes.
- Points Allocation — distribute a pool of points according to your personal preferences.
- Ranked Choice Voting — rank the list of options in order based on your personal preferences.
- MaxDiff Analysis — pick the best and worst options from a list of 3-6 choices.
- Conjoint Analysis — pick the best ‘profile’ (made up of a series of category-based variables) to identify the most important category and the ranked order of variables within each category.



## 2. Monadic Studies

Monadic studies are like the opposite of a comparative preference test. Rather than show options together, you completely isolate them by only showing one option at a time. Every option is assigned its own survey (each with an identical list of questions), which gets sent to a separate group of respondents. At the end, the results of all the surveys are aggregated and compared.



Unlike preference testing, monadic studies can't compare the main options against each other directly, so they generally rely on non-comparative question types such as:

- Likert scales → eg. "rate this mockup on a 5-point scale from disinterested to interested."
- Open-ended questions → eg. "what do you like most about the design of this product?"
- Semantic differential scales → eg. "rate this advert on a 5-point scale from budget to luxury."
- Brand perception → eg. "how would you describe the brand personality of this brand?"

- Price sensitivity → eg. "at what price point would you consider this product too expensive?"
- Recommendation → eg. "how likely are you to recommend [product] to a friend or colleague?"

The challenge with these non-comparative question formats is “**central tendency bias**”; an increased chance that participants give everything the same score, like rating everything “neutral” or “3/5 stars”.

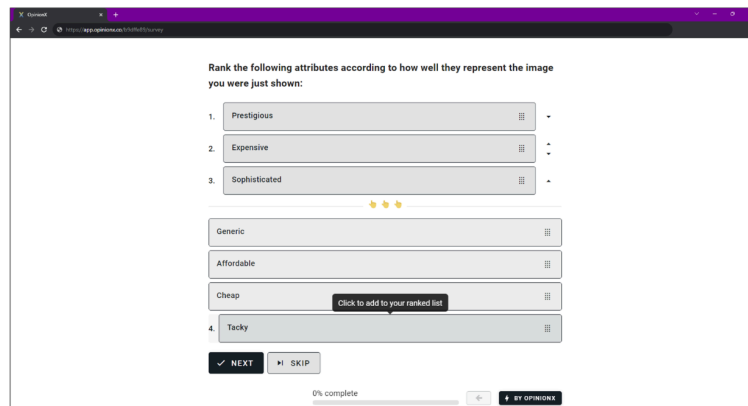
**Central Tendency Bias**

Data collection risk for non-comparative questions like rating scales

|                     | Very Dissatisfied        | Neutral                             | Very Satisfied           |
|---------------------|--------------------------|-------------------------------------|--------------------------|
| Salary              | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Job Perks           | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Career Growth       | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Healthcare Benefits | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Skill Development   | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Work Environment    | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

You can overcome central tendency bias by using comparison-based questions *within* your monadic studies. For example, a monadic study to understand consumer perception towards concept product designs could use a ranked choice voting question to show a list of brand attributes (like “luxurious”, “affordable”, “basic”) for participants to rank in relation to a concept image. This can help you

identify which version of your product is most associated with your target attribute/description like “luxurious”.

A screenshot of a web browser displaying a survey titled "Rank the following attributes according to how well they represent the image you were just shown:". The survey lists seven attributes: 1. Prestigious, 2. Expensive, 3. Sophisticated, 4. Generic, 5. Affordable, 6. Cheap, and 7. Tacky. Each attribute is in a box with a three-line menu icon on the right. The first three attributes are already ranked, with three yellow stars below them. A button labeled "Click to add to your ranked list" is positioned over the "Cheap" attribute. At the bottom, there are "NEXT" and "SKIP" buttons, a progress bar showing "0% complete", and a "57 OPTIONS" button.

Screenshot from a [Ranked Choice Voting](#) survey on [OpinionX](#)

### 3. Segmentation Analysis

Regardless of which A/B testing approach you decide to run — a single survey that uses comparative preference testing or a series of monadic studies — it is essential that you plan ahead for [segmentation analysis](#) afterwards.

**What is segmentation analysis?** It’s the process of filtering your results to only include participants that fall into a specific group of interest.

For example, you could filter your results to only view data from female respondents or you could compare the difference between male vs female respondents.

Segmentation is essential because representative samples lead to vague or average results. You don't want to know which option "people" pick more often — you need to know which option your most-ideal best-fit customers flock to. Here's an example of how segmentation can flip your results, showing how a low-performing option jumps to the best-performing one when viewed through the right segment:

## Conclusion

A/B testing is a popular type of experiment because it provides actionable data about people's behavior and how that translates to business outcomes. But sample size requirements often limit website A/B testing to large companies — that doesn't need to be the case! Using comparative preference testing methods like [pairwise comparison](#) or monadic studies will let you leverage the principles of A/B testing in much more flexible ways so that you can identify your best-performing image, feature idea, ad copy, sales messaging, problem statement, and more.

## **Web Search & Retrieval:**

In the digital age, the exponential growth of information has presented a remarkable challenge: how to efficiently access and retrieve the vast amount of knowledge available. Information retrieval (IR) serves as the vital framework that enables individuals to navigate this information landscape and unlock the treasures hidden within.

Web search and retrieval is the process of finding relevant documents, data, and web pages from a massive digital collection using a search query. It forms the core engine behind modern tools that fetch and rank internet content to match user intent.

### **Understanding Information Retrieval**

Information retrieval is the process of obtaining relevant information from a collection of resources, typically textual or multimedia documents, in response to a user's query. The primary goal is to bridge the gap between user queries and relevant information, empowering

individuals to find answers, explore new insights, and make informed decisions.

## Key Components of Information Retrieval

1. **Document Collection:** Information retrieval relies on a vast collection of documents, such as books, articles, websites, and multimedia content. These documents act as the foundation for the retrieval process.
2. **Indexing:** To facilitate efficient search and retrieval, documents are indexed. Indexing involves creating structured representations of the documents, capturing their key attributes, such as keywords, metadata, and semantic information. This process enables faster access to relevant information during retrieval.
3. **Query Processing:** When a user submits a query, the retrieval system analyzes the query to understand its intent. This involves parsing the query, identifying key terms, and formulating an effective search strategy to locate relevant documents.

4. **Ranking and Relevance:** Retrieval systems employ algorithms to rank the retrieved documents based on their relevance to the query. Various relevance metrics, such as term frequency, document popularity, and user feedback, are considered to ensure the most pertinent results are presented to the user. Once the data is matched, the system returns the relevant information to the user through a display or user interface.
5. **User Interface:** The user interface plays a crucial role in information retrieval.

## **Key Information Retrieval Techniques**

Let's take a look at the most common techniques in use today for information retrieval, their strengths and limitations, and real-world applications. Information retrieval systems often utilize a combination of these techniques to improve accuracy and efficiency.

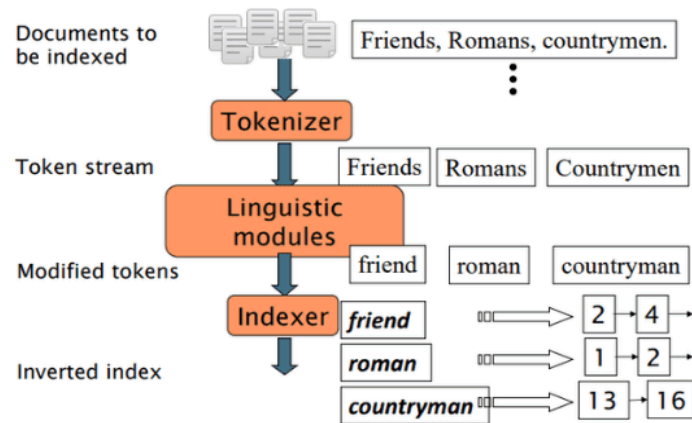
### **Boolean Retrieval**

One of the simplest methods, boolean retrieval involves retrieving information based on Boolean operations (AND, OR and NOT). This technique is an effective way to filter information and useful for its precision. A document is a match to the query or it is not. The Boolean method tends to be used by professionals who are searching for specific documents, such as legal searches.

While simple and efficient, a disadvantage of Boolean retrieval is there is no ranking of the results by how closely they match the needs of the user. There is also no nuance, meaning unless there is an exact match with the terms, the system will fail to retrieve results that share similarities with the query terms.

## Boolean Search: Harnessing AND, OR, and NOT Gates with Inverted Indexes

- Inverted Index
- Boolean Algebra
- Implementation
- Evaluation metrics



## Vector Space Model

In contrast, a vector space model (VSM) has the ability to capture nuances and retrieve results with semantic similarities to the query. In this algebraic method, documents and queries are represented as vectors in multidimensional space. Each vector embedding contains the semantic meanings of the documents or queries they represent. By measuring the distance or angle between the vectors, a user can

calculate the similarity between documents and queries (e.g., the similarity between “storm” and “rain”).

In this model, each term in a document earns a term weighting, which is a numerical value that reflects that term’s importance in the context of the document or query. Term weighting helps distinguish between common words, such as “and” and “the,” and rare words that might be more relevant.

Once the weights have been assigned, the cosine similarity, which measures the similarity between two vectors, determines a document’s relevance to a particular query. A higher cosine similarity indicates greater similarity between a document and query. The system then selects and ranks documents with the top scores.

In this way, the vector space model is useful for IR when it is important to retrieve and rank information that is the most relevant to the query. It’s widely used in search engines, enterprise search, recommendations, and e-commerce. It is also foundational in natural language processing (NLP) to take text data and make it suitable for NLP tasks.

## Documents



## Vector-space representation

However, complexity  
t We will see how small  
a Given a function-based  
a Using entropy of traffic  
a i  
t a  
l i  
f a  
r c  
r k  
s c  
r c  
s e  
r t  
n r  
s c  
u  
t

We study the complexity of influencing elections through bribery: How computationally complex is it for an external actor to determine whether by a certain amount of bribing voters a specified candidate can be made the election's winner? We study this problem for election systems as varied as scoring ...

|            | D1 | D2 | D3 | D4 | D5 |
|------------|----|----|----|----|----|
| complexity | 2  |    | 3  | 2  | 3  |
| algorithm  | 3  |    |    | 4  | 4  |
| entropy    | 1  |    |    | 2  |    |
| traffic    |    | 2  | 3  |    |    |
| network    |    | 1  | 4  |    |    |

Term-document matrix

# Vector Space Model

- Everything is a vector in some high-dimensional space
  - words, documents, queries, user preferences
- Issues to consider
  - what are the dimensions of that space (basis vectors)?
  - how to project words/documents/queries to that space?
  - how to compare documents and queries?
- Dimensions
  - basis: set of linearly-independent (orthogonal) vectors
  - “core” semantic concepts: works on toy datasets
  - words in the corpus: one dimension per distinct word
    - not orthogonal, huge dimensionality, constantly-growing

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**Probabilistic models:** Probabilistic models assess the likelihood of a document’s relevance to a query, ranking results based on this calculated probability. It calculates the probabilities using factors like how frequently a term appears in a document and document length.

These models are useful in situations where queries are not looking for an exact match or such a match is difficult to find. They are still widely used and foundational to systems today like search engines that benefit from the flexibility of ranking results by the likelihood of relevance.

### **Semantic search models**

Going beyond the simple keyword matching of traditional models, semantic search models get to the intent behind queries. They accomplish this through machine learning algorithms and natural language processing to understand the patterns and relationships in words.

These models have been shown to provide more accurate and relevant results than through keyword matching by understanding the context of queries, which leads to better search experiences for your employees.





